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1 Enterprise Intelligence and Investment Research Platform - Complete Software Development Plan

Document type: Developer-ready master requirements, architecture, scaling, and delivery plan

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Scope: Phase 1 U.S. MVP first; Phase 2 global expansion second.

Primary objective: Build a multi-user enterprise platform that turns public records, financial data, market data, government records, and document evidence into cited intelligence reports, investor analysis, and actionable monitoring workflows.

This document consolidates every major feature, data source, architecture proposal, database/lakehouse/warehouse strategy, Claude financial-skills integration, use case, API contract, developer backlog, acceptance criterion, and external URL discussed so far. It is intended to let a development team start implementation immediately.

1.1 Executive Summary

The product is a secure enterprise intelligence and investment research platform. It helps users investigate public companies, private companies, executives, investors, funds, government contractors, lobbyists, PACs, nonprofits, agencies, regulations, bills, contracts, grants, lawsuits, sanctions, and market data.

The system should feel like a combined equity research desk, investigative research team, public-records analyst, government affairs analyst, procurement analyst, legal-risk analyst, portfolio monitor, and report writer.

The central product promise is:

Enter a company, stock ticker, person, fund, agency, bill, regulation, contract, grant, court case, PAC, nonprofit, or portfolio. The platform resolves the entity, builds a cited intelligence graph, identifies related parties, analyzes investment implications, runs financial workflows, generates detailed reports, lets analysts edit and improve those reports in review, and monitors future changes.

Phase 1 is a U.S.-focused MVP that integrates official U.S. sources such as SEC EDGAR, FEC, LDA.gov, FARA, Congress.gov, Federal Register, Regulations.gov, eCFR, RegInfo/OIRA, USAspending, SAM.gov, IRS Form 990, CourtListener/RECAP, OFAC/OpenSanctions, OpenCorporates, GLEIF, and market-data providers.

Phase 2 expands the same ontology and evidence architecture globally, adding U.K., EU, Canada, Australia, Brazil, ICIJ, OCCRP/OpenAleph, OpenOwnership, World Bank procurement, global sanctions, and cross-border entity resolution.

1.2 Product Principles

- **Evidence first.** Every relationship, claim, risk flag, valuation data point, and report finding must trace to evidence.
- **Official source first.** Use official APIs and bulk data before scraping. Scraping is only for public, permitted, rate-limited pages when no API exists.
- **Human review for high-risk conclusions.** AI can suggest leads, but sensitive or investment-impacting conclusions require review and approval.
- **No unsupported allegations.** Reports should state what records show, label uncertainty, and avoid speculative accusations.
- **Investor decision support, not automatic trading.** The platform may provide research and structured analysis but should not execute trades or claim guaranteed outcomes.
- **Multi-user enterprise governance.** Every action should be permissioned, audited, versioned, and reversible where possible.
- **Reusable source contracts.** Every connector must have a source contract defining auth, limits, refresh cadence, compliance notes, raw storage, and normalized outputs.
- **Cost-aware data architecture.** Ingest and store only what creates product value; keep raw data cheap in object storage; materialize hot data only when needed.

1.3 Two-Phase Roadmap Overview

1.4 Phase 1 - U.S. MVP

Phase 1 creates a complete U.S.-first intelligence and investor research platform. MVP does not mean toy product. It means a focused first version with complete core workflows and U.S. sources only.

Phase 1 must deliver:

- Enterprise workspaces, users, roles, audit logs, projects, watchlists, and review workflows.
- U.S. public-record ingestion and evidence storage.
- Entity resolution and graph intelligence.
- Company and stock analysis.
- Claude financial-skills gateway for reports such as comps, DCF, earnings updates, one-pagers, due-diligence packs, and investment committee memos.
- Report generation, editing, enhancement, review, and export.
- Portfolio/watchlist monitoring and alerts.
- Cost-efficient storage, search, graph, and analytics stack.

1.5 Phase 2 - Global Expansion

Phase 2 extends the same platform to international ownership, procurement, sanctions, lobbying, political finance, corporate registries, and regulatory sources. Phase 2 should not fork the system. It should add jurisdiction-aware connectors and cross-border entity-resolution rules on top of Phase 1.

2 Phase 1 - U.S. MVP

2.1 Phase 1 Outcome

At the end of Phase 1, a user should be able to search a U.S. public company or private business, generate an evidence-backed entity profile, analyze the stock or company as an investor, discover related parties, view lobbying/regulatory/procurement/legal exposure, produce detailed reports, edit and enhance those reports during review, export them, and monitor future changes.

Example Phase 1 target flow:

1. User searches Palantir, PLTR, Peter Thiel, Anduril, or another U.S.-connected entity.
2. System resolves canonical entities using CIK, ticker, LEI, OpenCorporates, GLEIF, filings, aliases, and graph context.
3. System shows the entity profile: filings, officers, investors, holdings, lobbying, contracts, regulations, lawsuits, sanctions/exclusions, news/events, and related parties.
4. User selects Analyze Stock or Analyze Company.
5. System runs financial analysis, valuation, technical indicators, government exposure, lobbying exposure, regulatory exposure, legal risk, and related-party discovery.
6. User selects a report type: investor intelligence report, long thesis, short thesis, company intelligence report, procurement exposure report, lobbying report, or investment committee memo.
7. Platform generates a cited draft report.
8. Analyst edits the report, requests enhancements, adds or removes evidence, asks Claude finance skills to improve specific financial sections, and sends it to review.
9. Reviewer approves or rejects claims.
10. Final report exports to PDF, Word, PowerPoint, Excel model, Markdown, and evidence appendix.
11. User adds entity to watchlist and receives alerts for new filings, lobbying, contracts, regulations, market moves, and litigation.

2.2 Phase 1 Modules

Phase 1 contains the following modules. Each module must be built as a separately testable unit with clear APIs, permissions, and audit logs.

2.2.1 Enterprise Workspace and Identity

Organizations, workspaces, users, SSO-ready auth model, roles, permissions, project/case folders, audit logs, tenant isolation.

2.2.2 Source Registry and Connector Framework

Registry of sources, credentials, source contracts, ingestion runs, rate limits, retries, raw-data persistence, connector health dashboard.

2.2.3 Evidence Vault

Immutable raw documents, hashes, retrieval metadata, source records, evidence references, source snippets, document viewer.

2.2.4 U.S. Source Connectors

SEC, FEC, LDA, FARA, Congress.gov, GovInfo, Federal Register, Regulations.gov, eCFR, RegInfo/OIRA, USAspending, SAM.gov, IRS 990, CourtListener, OFAC/OpenSanctions, OpenCorporates, GLEIF.

2.2.5 Entity Resolution

Deterministic identifiers, fuzzy matching, graph-neighborhood matching, review queue, merge/unmerge, conflict rules.

2.2.6 Intelligence Graph

Entities, relationships, evidence-backed edges, pathfinding, related-party discovery, timelines, visual graph UI.

2.2.7 Investor Intelligence

Stock analysis, financial statements, valuation, DCF, comps, technical analysis, catalysts, insider/institutional data, government exposure, portfolio monitoring.

2.2.8 Claude Financial Skills Gateway

Adapters for Anthropic financial-services skills and agents: comps, DCF, earnings, one-pagers, investment committee memos, due diligence, valuation review, market research.

2.2.9 Report Generation and Review

Report templates, evidence bundles, AI drafting, citation verifier, review tasks, comments, redlines, edit/enhance workflow, exports.

2.2.10 Alerts and Monitoring

Watch entities, tickers, bills, dockets, agencies, contracts, filings, comments, lawsuits, sanctions, and market signals.

2.2.11 Admin, Compliance, and Governance

Source-policy registry, restricted-data handling, FEC/SAM controls, export approval, audit logs, retention policies.

2.3 Phase 1 Use Cases

2.3.1 Use Case: U.S. public-company investor report

An analyst enters a ticker. The platform resolves the company, pulls SEC filings, fundamentals, market data, lobbying records, contracts, regulations, litigation, and related parties. It generates a report with financials, valuation, catalysts, risks, and cited public-record findings.

2.3.2 Use Case: Day trader catalyst scan

A trader enters a ticker or watchlist. The system shows price/volume movement, volatility, pre-market/after-hours signals if licensed, recent filings, new contracts, new regulatory events, earnings events, news spikes, technical levels, and risk notes.

2.3.3 Use Case: Long-term investment thesis

A long-only investor runs a 12-36 month thesis report with business model, financial quality, DCF, peer comps, management, insider activity, government exposure, regulatory risks, and monitoring plan.

2.3.4 Use Case: Government contractor diligence

A user analyzes a contractor, linking USAspending awards, SAM opportunities/exclusions, agencies, subcontracting, lobbying, related entities, lawsuits, sanctions, and procurement concentration.

2.3.5 Use Case: Lobbying and regulatory exposure

A policy analyst enters a company, bill, agency, or docket. The platform maps clients, lobbying firms, lobbyists, issue codes, bills, agencies, Federal Register rules, Regulations.gov comments, eCFR sections, OIRA meetings, and affected companies.

2.3.6 Use Case: Related-party investigation

An investigator enters a person, fund, or company. The system discovers executives, board interlocks, investors, funds, PACs, nonprofits, lobbyists, law firms, contractors, court parties, shared addresses, registered agents, and public-record co-mentions.

2.3.7 Use Case: Portfolio monitoring

An investor uploads a portfolio. The platform maps tickers to entities, tracks filings, earnings, lobbying, contracts, regulations, litigation, market signals, risk events, and generates daily/weekly briefs.

2.3.8 Use Case: Report editing and enhancement

An analyst generates a draft report, edits sections, requests AI enhancement of selected paragraphs, adds evidence, regenerates financial sections via Claude financial skills, compares versions, and sends to review.

2.3.9 Use Case: Compliance/KYB public-record pack

A compliance team runs a public-record due-diligence pack on a vendor/counterparty, including corporate IDs, sanctions/exclusions, lawsuits, government contracts, ownership, officers, and risk flags.

2.4 Phase 1 Acceptance Criteria

- User can create an organization, workspace, project, watchlist, portfolio, and investigation case.
- User can search a U.S. company, ticker, person, agency, bill, docket, contract recipient, committee, PAC, nonprofit, or court case.
- System stores raw source records before normalization.
- Every normalized entity and relationship has at least one EvidenceRef.
- At least SEC, FEC, LDA.gov, Congress.gov, Federal Register, USAspending, OpenCorporates, GLEIF, CourtListener, and OpenSanctions/OFAC connectors are functioning in production mode.
- Additional connectors for FARA, Regulations.gov, eCFR, RegInfo/OIRA, SAM.gov, IRS 990, and GovInfo are available by the end of Phase 1 hardening.
- System supports deterministic entity matching and reviewable fuzzy matches.
- Graph UI displays evidence-backed edges and related-party suggestions.
- Stock analysis supports fundamentals, valuation, technical analysis, government exposure, and cited risk findings.
- Claude financial-skills gateway can run approved finance workflows through an adapter layer and store generated artifacts.
- Report editor supports comments, redlines/versioning, AI enhancement, claim review, citation verification, and export.
- Watchlists and alerts work for filings, lobbying, regulations, contracts, lawsuits, sanctions, and market events.
- Audit logs track searches, exports, report generation, claim approval, entity merges, source runs, and admin changes.

3 Phase 1 U.S. Source Connector Requirements

Every connector follows the same lifecycle:

1. Discover records.
2. Fetch records using official API or bulk data.
3. Save raw response/document to the evidence vault.
4. Parse source fields.

5. Normalize entities and identifiers.
6. Create evidence references.
7. Create candidate relationships.
8. Run entity resolution.
9. Store graph edges.
10. Index text and metadata.
11. Emit data-quality checks and source-run metrics.

Each connector must include:

- source_contract.yml
- typed Python client
- auth/secrets handling
- rate limiting
- retries with exponential backoff
- checkpointing
- idempotent ingestion
- raw-document storage
- parser tests
- fixture-based unit tests
- integration tests marked separately
- data quality checks
- source documentation
- compliance notes

3.1 SEC EDGAR / [data.sec.gov](https://www.sec.gov)

Primary URL: <https://www.sec.gov/search-filings/edgar-application-programming-interfaces>

Purpose: Public-company filings, XBRL facts, insider transactions, institutional holdings, beneficial ownership, private offerings, proxy relationships.

Required ingestion: - CIK lookup - ticker lookup - company submissions - company facts - 10-K - 10-Q - 8-K - S-1 - DEF 14A - Forms 3/4/5 - 13F - 13D - 13G - Form D

Normalized entities: - Company - Person - Fund - Security - Filing - InsiderTransaction - InstitutionalHolding - OwnershipStake - PrivateOffering

Core relationships: - Company FILED Filing - Person INSIDER_TRANSACTION Security - Fund HOLDS Security - Person OFFICER_OF Company - Person BOARD_MEMBER_OF Company

Developer implementation notes: - Add a typed client under packages/connectors/sec. - Save raw source responses before parsing. - Add deterministic IDs where available. - Emit SourceRun, SourceRecord, RawDocument, EvidenceRef, Entity, and Relationship records. - Add mocked fixtures and a small live integration test behind an environment flag. - Add data-

quality checks for missing IDs, duplicate source records, stale data, parse failures, and broken links.

3.2 FEC / OpenFEC

Primary URL: <https://api.open.fec.gov/developers/>

Purpose: Federal campaign finance, candidates, committees, PACs, contributions, disbursements, independent expenditures.

Required ingestion: - candidate search - committee search - receipts - disbursements - independent expenditures - candidate-committee links - bulk files

Normalized entities: - Person - Candidate - Committee - PAC - Contribution - Disbursement - IndependentExpenditure

Core relationships: - Person CONTRIBUTED_TO Committee - Committee SUPPORTS Candidate - Committee MADE_EXPENDITURE IndependentExpenditure

Developer implementation notes: - Add a typed client under packages/connectors/fec. - Save raw source responses before parsing. - Add deterministic IDs where available. - Emit SourceRun, SourceRecord, RawDocument, EvidenceRef, Entity, and Relationship records. - Add mocked fixtures and a small live integration test behind an environment flag. - Add data-quality checks for missing IDs, duplicate source records, stale data, parse failures, and broken links.

3.3 LDA.gov Lobbying API

Primary URL: <https://lda.gov/api/redoc/v1/>

Purpose: Federal lobbying registrations, quarterly reports, contribution reports, clients, registrants, lobbyists, issues, bills, agencies.

Required ingestion: - LD-1 registrations - LD-2 reports - LD-203 reports - clients - registrants - lobbyists - issue codes - agencies - bills - amounts

Normalized entities: - Company - LobbyingFirm - Lobbyist - LobbyingReport - Agency - Bill - Issue

Core relationships: - Company LOBBIED_ON Bill - Company LOBBIED_AGENCY Agency - Lobbyist REPRESENTED Client - LobbyingFirm FILED LobbyingReport

Developer implementation notes: - Add a typed client under packages/connectors/ldagov. - Save raw source responses before parsing. - Add deterministic IDs where available. - Emit SourceRun, SourceRecord, RawDocument, EvidenceRef, Entity, and Relationship records. - Add mocked fixtures and a small live integration test behind an environment flag. - Add data-quality checks for missing IDs, duplicate source records, stale data, parse failures, and broken links.

3.4 FARA / DOJ

Primary URL: <https://efile.fara.gov/ords/fara/f?p=1235:10>

Purpose: Foreign principals, foreign agents, registrants, political activities, informational materials, receipts/disbursements.

Required ingestion: - registrants - foreign principals - filings - supplemental statements - informational materials - political activities

Normalized entities: - ForeignPrincipal - Registrant - FARAfiling - Organization - Person - Agency

Core relationships: - ForeignPrincipal HIRED Registrant - Registrant FILED FARAfiling - FARAfiling MENTIONS Agency

Developer implementation notes: - Add a typed client under packages/connectors/fara. - Save raw source responses before parsing. - Add deterministic IDs where available. - Emit SourceRun, SourceRecord, RawDocument, EvidenceRef, Entity, and Relationship records. - Add mocked fixtures and a small live integration test behind an environment flag. - Add data-quality checks for missing IDs, duplicate source records, stale data, parse failures, and broken links.

3.5 Congress.gov

Primary URL: <https://api.congress.gov/>

Purpose: Bills, amendments, members, sponsors, cosponsors, committees, legislative actions.

Required ingestion: - bills - summaries - sponsors - cosponsors - actions - committees - amendments - members

Normalized entities: - Bill - Legislator - Committee - Action - Amendment

Core relationships: - Legislator SPONSORED Bill - Legislator COSPONSORED Bill - Bill REFERRED_TO Committee

Developer implementation notes: - Add a typed client under packages/connectors/congressgov. - Save raw source responses before parsing. - Add deterministic IDs where available. - Emit SourceRun, SourceRecord, RawDocument, EvidenceRef, Entity, and Relationship records. - Add mocked fixtures and a small live integration test behind an environment flag. - Add data-quality checks for missing IDs, duplicate source records, stale data, parse failures, and broken links.

3.6 GovInfo

Primary URL: <https://api.govinfo.gov/docs/>

Purpose: Official federal document packages including Congressional Record, hearings, bills, committee reports, statutes, CFR materials.

Required ingestion: - packages - granules - metadata - PDFs - text - hearings - committee reports

Normalized entities: - OfficialDocument - Hearing - Report - Bill - CongressionalRecordItem

Core relationships: - Document CITES Bill - Hearing DISCUSSES Agency - Report RELATES_TO Bill

Developer implementation notes: - Add a typed client under packages/connectors/govinfo. - Save raw source responses before parsing. - Add deterministic IDs where available. - Emit SourceRun, SourceRecord, RawDocument, EvidenceRef, Entity, and Relationship records. - Add mocked fixtures and a small live integration test behind an environment flag. - Add data-quality checks for missing IDs, duplicate source records, stale data, parse failures, and broken links.

3.7 Federal Register

Primary URL: <https://www.federalregister.gov/developers/documentation/api/v1>

Purpose: Proposed rules, final rules, notices, executive orders, agency actions, RINs, docket references.

Required ingestion: - documents - agencies - proposed rules - final rules - notices - presidential docs - CFR refs - docket IDs - RINs

Normalized entities: - Rule - Notice - Agency - RIN - CFRSection - Docket

Core relationships: - Agency ISSUED Rule - Rule AMENDS CFRSection - Rule RELATES_TO Docket

Developer implementation notes: - Add a typed client under packages/connectors/federal. - Save raw source responses before parsing. - Add deterministic IDs where available. - Emit SourceRun, SourceRecord, RawDocument, EvidenceRef, Entity, and Relationship records. - Add mocked fixtures and a small live integration test behind an environment flag. - Add data-quality checks for missing IDs, duplicate source records, stale data, parse failures, and broken links.

3.8 Regulations.gov

Primary URL: <https://open.gsa.gov/api/regulationsgov/>

Purpose: Dockets, documents, public comments, attachments, supporting materials.

Required ingestion: - dockets - documents - comments - attachments - submitters - organizations - posted dates - comment text

Normalized entities: - Docket - PublicComment - Document - Attachment - Organization - Agency

Core relationships: - Organization SUBMITTED_COMMENT PublicComment - PublicComment RELATES_TO Docket - Docket MANAGED_BY Agency

Developer implementation notes: - Add a typed client under packages/connectors/regulationsgov. - Save raw source responses before parsing. - Add deterministic IDs where available. - Emit SourceRun, SourceRecord, RawDocument, EvidenceRef, Entity, and Relationship records. - Add mocked fixtures and a small live integration test behind an environment flag. - Add data-quality checks for missing IDs, duplicate source records, stale data, parse failures, and broken links.

3.9 eCFR

Primary URL: <https://www.ecfr.gov/developers/documentation/api/v1>

Purpose: Current and historical regulatory text, title/part/section retrieval, text diffs.

Required ingestion: - titles - parts - sections - XML/text - historical versions - recent changes

Normalized entities: - CFRTitle - CFRPart - CFRSection - RegulatoryTextVersion

Core relationships: - Rule AMENDS CFRSection - CFRSection VERSIONED_AS RegulatoryTextVersion

Developer implementation notes: - Add a typed client under packages/connectors/ecfr. - Save raw source responses before parsing. - Add deterministic IDs where available. - Emit SourceRun, SourceRecord, RawDocument, EvidenceRef, Entity, and Relationship records. - Add mocked fixtures and a small live integration test behind an environment flag. - Add data-quality checks for missing IDs, duplicate source records, stale data, parse failures, and broken links.

3.10 RegInfo/OIRA/Unified Agenda

Primary URL: <https://www.reginfo.gov/public/>

Purpose: Regulatory pipeline, RINs, OIRA reviews, meeting participants, Unified Agenda items.

Required ingestion: - Unified Agenda - RINs - rule stages - OIRA reviews - OIRA meetings - participants

Normalized entities: - RIN - OIRAMeeting - Rule - Agency - Organization - Person

Core relationships: - Organization ATTENDED_OIRA_MEETING OIRAMeeting - OIRAMeeting RELATES_TO RIN - Agency REVIEWED Rule

Developer implementation notes: - Add a typed client under packages/connectors/reginfo_oira_unified. - Save raw source responses before parsing. - Add deterministic IDs where available. - Emit SourceRun, SourceRecord, RawDocument, EvidenceRef, Entity, and Relationship records. - Add mocked fixtures and a small live integration test behind an environment flag. - Add data-quality checks for missing IDs, duplicate source records, stale data, parse failures, and broken links.

3.11 USAspending

Primary URL: <https://api.usaspending.gov/>

Purpose: Federal contracts, grants, loans, direct payments, recipients, agencies, subawards.

Required ingestion: - awards - recipients - agencies - subawards - NAICS - PSC - award amounts - award dates - recipient IDs

Normalized entities: - Award - Recipient - Agency - Program - Subaward - Company

Core relationships: - Company RECEIVED_CONTRACT Award - Company RECEIVED_GRANT Award - Award FUNDED_BY Agency

Developer implementation notes: - Add a typed client under packages/connectors/usaspending. - Save raw source responses before parsing. - Add deterministic IDs where available. - Emit SourceRun, SourceRecord, RawDocument, EvidenceRef, Entity, and Relationship records. - Add mocked fixtures and a small live integration test behind an environment flag. - Add data-quality checks for missing IDs, duplicate source records, stale data, parse failures, and broken links.

3.12 SAM.gov

Primary URL: <https://sam.gov/>

Purpose: Contract opportunities, public entity data, exclusions, assistance listings.

Required ingestion: - opportunities - entities - exclusions - NAICS - PSC - UEI - CAGE if public/allowed - solicitation IDs

Normalized entities: - Opportunity - Vendor - Exclusion - Agency - AssistanceListing

Core relationships: - Vendor ELIGIBLE_FOR Opportunity - Vendor EXCLUDED_BY Exclusion - Agency POSTED Opportunity

Developer implementation notes: - Add a typed client under packages/connectors/samgov. - Save raw source responses before parsing. - Add deterministic IDs where available. - Emit SourceRun, SourceRecord, RawDocument, EvidenceRef, Entity, and Relationship records. - Add mocked fixtures and a small live integration test behind an environment flag. - Add data-quality checks for missing IDs, duplicate source records, stale data, parse failures, and broken links.

3.13 IRS Form 990

Primary URL: <https://www.irs.gov/charities-non-profits/form-990-series-downloads>

Purpose: Nonprofits, foundations, think tanks, officers, grants, grantees, related organizations.

Required ingestion: - Form 990 - Form 990-PF - EIN - officers - grants - grantees - related organizations - compensation

Normalized entities: - Nonprofit - Foundation - Grant - Person - Organization

Core relationships: - Foundation GRANTED_TO Organization - Person OFFICER_OF Nonprofit - Nonprofit RELATED_TO Organization

Developer implementation notes: - Add a typed client under packages/connectors/irs. - Save raw source responses before parsing. - Add deterministic IDs where available. - Emit SourceRun, SourceRecord, RawDocument, EvidenceRef, Entity, and Relationship records. - Add mocked fixtures and a small live integration test behind an environment flag. - Add data-quality checks for missing IDs, duplicate source records, stale data, parse failures, and broken links.

3.14 CourtListener / RECAP

Primary URL: <https://wiki.free.law/c/courtlistener/help/api/rest/v4/overview>

Purpose: Court opinions, dockets, RECAP filings, parties, attorneys, judges, legal-risk monitoring.

Required ingestion: - opinions - dockets - docket entries - RECAP metadata - parties - attorneys - judges

Normalized entities: - CourtCase - Docket - Opinion - Party - Attorney - LawFirm - Judge

Core relationships: - Company PARTY_TO Case - Person PARTY_TO Case - LawFirm REPRESENTED Party

Developer implementation notes: - Add a typed client under packages/connectors/courtlistener. - Save raw source responses before parsing. - Add deterministic IDs where available. - Emit SourceRun, SourceRecord, RawDocument, EvidenceRef, Entity, and Relationship records. - Add mocked fixtures and a small live integration test behind an environment flag. - Add data-quality checks for missing IDs, duplicate source records, stale data, parse failures, and broken links.

3.15 OFAC / OpenSanctions / SAM Exclusions

Primary URL: <https://github.com/opensanctions/opensanctions>

Purpose: Sanctions, PEPs, exclusions, risk flags, watchlist screening.

Required ingestion: - OFAC SDN - non-SDN - OpenSanctions exports/API - SAM exclusions - aliases - programs - dates

Normalized entities: - SanctionsRecord - ExclusionRecord - Person - Company - RiskFlag

Core relationships: - Entity LISTED_ON SanctionsRecord - Company EXCLUDED_BY ExclusionRecord

Developer implementation notes: - Add a typed client under packages/connectors/ofac. - Save raw source responses before parsing. - Add deterministic IDs where available. - Emit SourceRun, SourceRecord, RawDocument, EvidenceRef, Entity, and Relationship records. -

Add mocked fixtures and a small live integration test behind an environment flag. - Add data-quality checks for missing IDs, duplicate source records, stale data, parse failures, and broken links.

3.16 OpenCorporates

Primary URL: <https://api.opencorporates.com/documentation/API-Reference>

Purpose: Corporate registry lookup, company numbers, jurisdictions, officers, branches, registry links.

Required ingestion: - company search - profile - officers - registry IDs - jurisdictions - addresses

Normalized entities: - RegistryCompany - Company - Officer - Address - Jurisdiction

Core relationships: - Company SAME_AS RegistryCompany - Person OFFICER_OF Company

Developer implementation notes: - Add a typed client under packages/connectors/opencorporates. - Save raw source responses before parsing. - Add deterministic IDs where available. - Emit SourceRun, SourceRecord, RawDocument, EvidenceRef, Entity, and Relationship records. - Add mocked fixtures and a small live integration test behind an environment flag. - Add data-quality checks for missing IDs, duplicate source records, stale data, parse failures, and broken links.

3.17 GLEIF / LEI

Primary URL: <https://www.gleif.org/en/lei-data/gleif-api>

Purpose: LEI lookup, legal entity normalization, parent/child relationships, entity identity resolution.

Required ingestion: - LEI search - legal name - registered address - headquarters - parent relationships - status

Normalized entities: - LegalEntity - LEI - Company - Address

Core relationships: - Company HAS_LEI LEI - Company PARENT_OF Company - Company SAME_AS LegalEntity

Developer implementation notes: - Add a typed client under packages/connectors/gleif. - Save raw source responses before parsing. - Add deterministic IDs where available. - Emit SourceRun, SourceRecord, RawDocument, EvidenceRef, Entity, and Relationship records. - Add mocked fixtures and a small live integration test behind an environment flag. - Add data-quality checks for missing IDs, duplicate source records, stale data, parse failures, and broken links.

4 Phase 1 Feature Requirements

4.1 Enterprise Workspace and Multi-User System

4.2 Functional requirements

- Multi-tenant organizations.
- Workspaces per client, fund, team, or research unit.
- Project and case folders.
- Shared watchlists.
- Shared reports.
- Shared portfolios.
- User invitations.
- Guest reviewers.
- Role-based access control.
- Attribute-based access control for sensitive sources and exports.
- SSO-ready architecture with SAML/OIDC hooks.
- SCIM-ready provisioning model for enterprise rollout.
- Full audit logs.
- Admin dashboards for usage, sources, connectors, billing hooks, and security settings.

4.3 Required roles

- Organization owner.
- Workspace admin.
- Senior analyst.
- Analyst.
- Investor user.
- Compliance reviewer.
- Guest reviewer.
- Service account.

4.4 Permissions must apply to

- Workspaces.
- Projects/cases.
- Sources.
- Entities.
- Reports.
- Portfolios.
- Watchlists.
- Exports.
- AI runs.
- Entity merges.
- Claim approvals.

- Admin settings.

4.5 Evidence Vault and Source Registry

The evidence vault is the most important trust layer. It stores raw source documents and provides durable proof for every claim.

4.6 Required objects

```
{
  "Source": "metadata about an API, bulk dataset, or public source",
  "SourceCredential": "workspace-specific credential reference",
  "SourceContract": "auth, limits, refresh cadence, compliance notes, raw storage requirements",
  "SourceRun": "one ingestion job with status, timestamps, counts, errors",
  "SourceRecord": "one source-level record or API object",
  "RawDocument": "immutable raw JSON/XML/HTML/PDF/CSV/text object with hash",
  "EvidenceRef": "specific source field, excerpt, or document span supporting a normalized fact"
}
```

4.7 Required fields for every evidence reference

```
{
  "source_name": "SEC EDGAR",
  "source_url": "https://...",
  "source_record_id": "source-native ID or generated stable ID",
  "document_type": "filing/api_response/pdf/comment/docket/contract",
  "retrieved_at": "2026-05-25T00:00:00Z",
  "published_at": "source publication date if available",
  "filing_date": "filing date if available",
  "raw_document_hash": "sha256",
  "extraction_method": "official_api | bulk_data | permitted_scrape | manual_upload | licensed_vendor",
  "evidence_excerpt": "short field or excerpt",
  "confidence_score": 0.95
}
```

4.8 Entity Resolution and Graph Intelligence

4.9 Deterministic identifiers

The system should match exactly on deterministic identifiers when available:

- CIK.
- LEI.
- EIN.

- UEI.
- CAGE.
- FEC candidate ID.
- FEC committee ID.
- LDA registrant/client ID.
- FARA registrant number.
- OpenCorporates jurisdiction + company number.
- CourtListener docket ID.
- Congress.gov bill ID.
- Regulations.gov docket ID.
- RIN.
- USAspending recipient ID.
- OFAC/OpenSanctions IDs.

4.10 Probabilistic features

- Name similarity.
- Alias similarity.
- Address similarity.
- Officer overlap.
- Registered-agent overlap.
- Parent/subsidiary overlap.
- Same domains or websites where public.
- Same phone/email only when business-public and necessary.
- Co-occurrence in filings.
- Temporal plausibility.
- Jurisdiction plausibility.
- Graph-neighborhood similarity.

4.11 Merge policy

- High-confidence deterministic matches can auto-merge.
- Medium-confidence matches go to analyst review.
- Low-confidence matches create POSSIBLE_MATCH edges.
- Conflicting identifiers block auto-merge.
- Every merge must be reversible.
- Entity history must preserve all source aliases and identifiers.

4.12 Related-Party Discovery

The related-party engine finds other entities that may matter to an investor, analyst, or investigator.

4.13 Input parameters

```
{
  "entity_id": "uuid",
  "max_depth": 3,
  "relationship_types": ["OFFICER_OF", "BOARD_MEMBER_OF", "INVESTED_IN",
"LOBBIED_FOR", "RECEIVED_CONTRACT", "PARTY_TO_CASE"],
  "source_filters": ["SEC", "LDA", "USAspending", "CourtListener"],
  "date_range": {"from": "2020-01-01", "to": "2026-05-25"},
  "min_confidence": 0.65,
  "include_possible_matches": false,
  "ranking_mode": "investor_relevance|risk_relevance|influence_relevance|general"
}
```

4.14 Ranking signals

- Relationship strength.
- Evidence quality.
- Source authority.
- Recency.
- Frequency.
- Amount of money involved.
- Graph centrality.
- Cross-source corroboration.
- Relevance to requested report type.
- Regulatory/procurement/legal importance.

4.15 Output fields

```
{
  "related_entity_id": "uuid",
  "name": "Example Lobbying Firm",
  "type": "LobbyingFirm",
  "relationship_summary": "Represented Company X in 2025 LDA filings",
  "relationship_paths": [],
  "confidence": 0.91,
  "first_observed": "2025-01-01",
  "last_observed": "2025-12-31",
  "why_it_matters": "Shows federal policy activity on defense procurement issues",
  "evidence_refs": []
}
```

4.16 Analyze Stock / Analyze Company

This is the core investor feature. It combines market data, fundamentals, SEC filings, public-record intelligence, related-party discovery, and Claude financial skills.

4.17 Input schema

```
{  
  "query": "PLTR or Palantir",  
  "analysis_mode": "day_trader | swing_trader | long_term_investor | value_investor |  
growth_investor | dividend_investor | event_driven | activist | short_thesis | credit |  
private_equity | government_contracting | regulatory_risk",  
  "time_horizon": "intraday | 1-5_days | 1-3_months | 12-36_months | 5_years",  
  "risk_profile": "conservative | moderate | aggressive",  
  "portfolio_context_id": "optional uuid",  
  "peer_group": ["optional tickers"],  
  "include_modules": [  
    "fundamentals", "valuation", "dcf", "comps", "technicals", "earnings", "insiders",  
"institutional_ownership", "lobbying", "contracts", "regulation", "litigation", "sanctions",  
"related_parties", "news", "catalysts"  
  ],  
  "output_format": "dashboard | report | memo | excel_model | powerpoint"  
}
```

4.18 Required output sections

- Company overview.
- Stock snapshot.
- Price trend and volume trend.
- Liquidity and volatility.
- Fundamental summary.
- Valuation summary.
- DCF assumptions and fair-value range.
- Peer comps.
- Technical analysis.
- Earnings summary.
- Insider transactions.
- Institutional ownership.
- Government exposure.
- Lobbying exposure.
- Regulatory exposure.
- Procurement exposure.
- Litigation/enforcement risk.
- Sanctions/exclusions check.
- Related parties.
- Catalysts.
- Bull/base/bear cases.
- Risk factors.
- Monitoring plan.

- Evidence appendix.

4.19 Technical analysis indicators

- Moving averages.
- VWAP.
- RSI.
- MACD.
- Bollinger Bands.
- ATR.
- Support/resistance.
- Relative volume.
- Trend classification.
- Gap analysis.
- Momentum score.
- Volatility regime.

4.20 Valuation requirements

- Trading multiples.
- Peer comps.
- Historical multiple range.
- DCF.
- Reverse DCF.
- Sum-of-the-parts where applicable.
- Scenario analysis.
- Sensitivity tables.
- Bull/base/bear valuation.
- Assumption log.

4.21 Important output policy

The system should not produce unsupported buy, sell, or guaranteed return conclusions. It should produce structured research: setup quality, thesis, risks, assumptions, scenarios, evidence, and monitoring triggers.

4.22 Claude Financial Skills Gateway

Anthropic's public financial-services materials describe reference agents, skills, and data connectors for investment banking, equity research, private equity, and wealth management workflows. Phase 1 should integrate these through a provider-agnostic FinanceSkillGateway, not by hard-coding report logic to one vendor.

4.23 Gateway purpose

- Run comparable company analysis.
- Build DCF valuation models.

- Generate post-earnings reports.
- Generate one-page company profiles.
- Draft investment committee memos.
- Generate due-diligence data packs.
- Review valuation assumptions.
- Generate financial proposal sections.
- Enhance investor reports inside the review editor.

4.24 Adapter design

```
packages/skills_gateway/
registry.py
providers/
  anthropic_financial_services.py
  internal_finance.py
  mock_provider.py
skills/
  comps.py
  dcf.py
  earnings.py
  one_pager.py
  ic_memo.py
  due_diligence.py
  model_review.py
  market_research.py
security/
  sandbox.py
  allowlist.py
  audit.py
```

4.25 Skill invocation schema

```
{
  "provider": "anthropic_financial_services",
  "skill": "dcf",
  "entity_id": "uuid",
  "ticker": "PLTR",
  "input_bundle_id": "evidence_bundle_uuid",
  "assumptions": {
    "revenue_growth": [0.25, 0.22, 0.18, 0.15, 0.12],
    "terminal_growth": 0.03,
    "discount_rate": 0.10,
    "margin_path": "user_defined_or_model_suggested"
  },
  "output_artifacts": ["markdown", "excel", "chart_data"],
```

```
"review_required": true
}
```

4.26 Skill output schema

```
{
  "skill_run_id": "uuid",
  "status": "completed | failed | needs_review",
  "summary": "DCF suggests fair value range based on stated assumptions",
  "artifacts": [
    {"type": "markdown", "url": "object-storage-url"},
    {"type": "xlsx", "url": "object-storage-url"}
  ],
  "assumptions_used": {},
  "evidence_refs": [],
  "warnings": ["Outputs are research support and require analyst review"],
  "created_claims": []
}
```

4.27 Security controls

- Approved skills allowlist.
- Version-pinned skill packages.
- Sandboxed execution.
- No arbitrary network access by default.
- No unrestricted file access.
- Full invocation logs.
- Source/evidence bundle attached to every run.
- Human review required for published investment outputs.
- No automatic trading or broker execution from skill output.

4.28 Report Creation, Editing, Enhancement, and Review

The report system must support a full analyst workflow, not just one-shot generation.

4.29 Report lifecycle

1. User chooses report type and scope.
2. System builds an evidence bundle.
3. AI drafts report sections from evidence.
4. Claim verifier checks every factual statement.
5. Report opens in an editor with source sidebar.
6. Analyst edits text directly.
7. Analyst can highlight a paragraph and choose Enhance, Make more concise, Add financial detail, Add citations, Rewrite for investment committee, Rewrite for compliance, Add bear case, or Add source-backed evidence.

8. Analyst can run Claude financial skills on selected sections, such as DCF, comps, earnings, or IC memo.
9. All AI changes are shown as suggestions/redlines.
10. Analyst accepts/rejects suggestions.
11. Reviewer approves claims and language.
12. Final report exports with evidence appendix.

4.30 Report editing parameters

```
{  
  "report_id": "uuid",  
  "section_id": "uuid",  
  "operation": "enhance | rewrite | summarize | expand | add_citations | add_risk_factors |  
update_financials | regenerate_section",  
  "tone": "institutional | executive | technical | compliance | investor_memo | board_brief",  
  "evidence_scope": "current_section | whole_report | selected_evidence | new_retrieval",  
  "model_provider": "openai | anthropic | internal",  
  "finance_skill": "optional comps | dcf | earnings | ic_memo",  
  "review_required": true  
}
```

4.31 Report types in Phase 1

- Stock analysis report.
- Long thesis report.
- Short thesis report.
- Earnings preview.
- Earnings reaction report.
- DCF valuation report.
- Comparable-company report.
- Investment committee memo.
- Company intelligence report.
- Executive/person profile.
- Related-party report.
- Lobbying influence report.
- Regulatory exposure report.
- Government procurement report.
- Political-money report.
- Court/litigation report.
- Sanctions/exclusions report.
- Vendor/counterparty due diligence pack.
- Portfolio risk report.

4.32 Export formats

- PDF.
- Word.
- PowerPoint.
- Excel model.
- Markdown.
- HTML.
- JSON.
- CSV evidence appendix.

4.33 Alerts and Portfolio Monitoring

4.34 Alert input schema

```
{  
  "watch_target_type": "entity | ticker | portfolio | bill | docket | agency | keyword | source",  
  "watch_target_id": "uuid or string",  
  "trigger_types": ["new_sec_filing", "new_lobbying_report", "new_contract_award",  
"new_rule", "new_comment", "new_court_case", "sanctions_hit", "price_move",  
"volume_spike", "earnings_event"],  
  "thresholds": {"price_move_pct": 5, "contract_min_value": 1000000},  
  "delivery_channels": ["in_app", "email", "slack", "teams", "webhook"],  
  "frequency": "realtime | hourly | daily_digest | weekly_digest"  
}
```

4.35 Portfolio features

- CSV upload.
- Manual positions.
- Watchlists.
- Position-to-entity mapping.
- Cost basis.
- Unrealized P/L.
- Sector exposure.
- Factor exposure.
- Regulatory exposure.
- Government-contract exposure.
- Litigation and sanctions risk exposure.
- Earnings calendar.
- Catalyst calendar.
- Portfolio memo.
- Thesis tracker.

5 Phase 2 - Global Expansion

5.1 Phase 2 Outcome

Phase 2 adds global coverage without changing the core product model. The same evidence vault, ontology, entity resolution, graph, report editor, and finance modules should work across jurisdictions.

Global expansion should focus on sources that help answer questions such as:

- Does a U.S. company have foreign subsidiaries, beneficial owners, officers, contracts, or sanctions exposure?
- Is a foreign principal lobbying or influencing U.S. policy?
- Do international procurement or subsidies affect a company's revenue or competitive position?
- Are EU, U.K., Canada, Australia, Brazil, or other regulatory changes relevant to U.S. companies or portfolios?
- Are offshore entities, cross-border corporate structures, or PEP/sanctions links present?

5.2 Phase 2 Priority Modules

5.3 Global entity and jurisdiction layer

- Jurisdiction codes.
- Registry-specific identifiers.
- Multilingual names and aliases.
- Transliteration handling.
- Address normalization by country.
- Local currency handling.
- Local date formats.
- International source confidence rules.
- Cross-border same-as matching.

5.4 International corporate registries

- U.K. Companies House.
- EU/member-state registries where accessible.
- Canada corporate registries where accessible.
- Australia ASIC.
- Brazil corporate/market sources.
- OpenCorporates as broad discovery.
- GLEIF as identifier spine.

5.5 International lobbying and political finance

- EU Transparency Register.
- Canada lobbying open data.

- Australia lobbyist register.
- U.K. Register of Consultant Lobbyists where relevant.
- Brazil e-Agendas and political/election sources.

5.6 International procurement and grants

- EU TED.
- U.K. Contracts Finder / Find a Tender.
- CanadaBuys.
- AusTender.
- Brazil Portal da Transparencia and Compras.gov.br.
- World Bank procurement.
- UNGM.

5.7 International sanctions and risk

- UN sanctions.
- EU sanctions map.
- U.K. OFSI sanctions.
- OpenSanctions global enrichment.
- World Bank debarment.
- Multilateral development bank exclusions.

5.8 Offshore and investigative sources

- ICIJ Offshore Leaks.
- OCCRP Aleph / OpenAleph.
- OpenOwnership / BODS where available.
- FollowTheMoney-compatible datasets.

5.9 Phase 2 Acceptance Criteria

- Global connectors produce the same Entity, Identifier, Relationship, EvidenceRef, and SourceRecord structures used in Phase 1.
- Reports can distinguish U.S. and international evidence by jurisdiction and source reliability.
- Cross-border related-party discovery works through LEI, OpenCorporates, local registry IDs, directors, addresses, sanctions IDs, and corporate parent relationships.
- Global reports include currency conversion assumptions and source-date cutoffs.
- Multilingual sources can be stored in original language and translated for analyst review.
- Sanctions/PEP matches are evidence-backed and confidence-scored, not treated as automatic conclusions.

6 Architecture and Development Blueprint

6.1 Recommended Phase 1 Stack

6.2 Application layer

- Backend: Python + FastAPI.
- Frontend: Next.js + TypeScript.
- Worker jobs: Celery/RQ or Temporal; Prefect/Dagster for data assets.
- Auth: start with app-native auth; design for SAML/OIDC/SCIM.
- API style: REST for core app; internal async jobs for long-running ingest/report tasks.

6.3 Storage layer

- Postgres for operational system of record.
- Object storage for raw evidence and generated artifacts.
- OpenSearch for document search and later vector/hybrid search.
- pgvector for lightweight embeddings in Phase 1 if separate vector infra is not justified.
- Postgres edge tables for graph MVP.
- Optional Neo4j/ArangoDB only after graph workloads exceed Postgres practicality.

6.4 Analytics layer

- Start with object storage + Parquet + DuckDB for low-cost analytics.
- Add ClickHouse when event/relationship analytics and dashboard latency become heavy.
- Add Iceberg/Trino/Athena or managed warehouse when data volume, concurrency, and team workflows justify it.

6.5 Repository Structure

enterprise-intelligence-platform/

AGENTS.md

README.md

docker-compose.yml

pyproject.toml

package.json

apps/

api/

web/

admin/

worker/

packages/

connectors/

sec_edgar/

fec/
lda/
fara/
congress/
govinfo/
federal_register/
regulations_gov/
ecfr/
reginfo_oira/
usaspending/
sam_gov/
irs_990/
courtlister/
sanctions/
opencorporates/
gleif/
market_data/
ontology/
entity_resolution/
graph/
search/
evidence/
reporting/
finance/
skills_gateway/
alerts/
portfolios/
compliance/
exports/

infra/
docker/
kubernetes/
terraform/
migrations/

docs/
architecture/
source_contracts/
methodology/
report_templates/
compliance/
security/

tests/
unit/
integration/
fixtures/
contract/

6.6 Core Services

api-service	Public/internal API for app features.
web-app	Next.js user interface.
admin-service	Admin, billing hooks, source and user management.
worker-service	Ingestion, parsing, report jobs, alert scans.
connector-service	Source-specific connector execution.
evidence-service	Raw document storage, retrieval, hashing, evidence refs.
entity-resolution-service	Matching, merge queue, conflicts, review.
graph-service	Relationship queries, paths, related-party discovery.
search-service	Full-text, semantic, and entity search.
finance-service	Stock analysis, financial statements, valuation, technicals.
skills-gateway-service	Claude finance skills and internal skill adapters.
report-service	Evidence bundles, claim generation, citation verification, exports.
alert-service	Watchlists, alert rules, notifications.
compliance-service	Policy enforcement, restricted data, high-risk language review.
audit-service	Audit events and export logs.

7 Database, Data Lake, Warehouse, and Cost Strategy

7.1 Core Principle

Do not put everything into the most expensive database. Use separate storage for different workloads:

- Operational queries: Postgres.
- Raw evidence: object storage.
- Full-text document search: OpenSearch/Elasticsearch.
- Graph MVP: Postgres edge tables.
- Heavy graph analytics: Neo4j/ArangoDB/Neptune later.
- Cheap analytics: Parquet + DuckDB.
- Large-scale lakehouse: Iceberg + object storage + Trino/Athena/Spark.
- Real-time OLAP dashboards: ClickHouse.
- Enterprise warehouse: BigQuery/Snowflake/Databricks when needed.

7.2 Proposal A - Low-Cost Phase 1 Stack

7.3 Recommended for immediate MVP

Use:

- Postgres for app database, entities, identifiers, relationships, reports, claims, users, alerts.
- S3-compatible object storage for raw JSON/XML/PDF/HTML/CSV and report artifacts.
- Parquet export jobs for large source tables.
- DuckDB for analyst-style batch queries over Parquet.
- OpenSearch for filings, comments, dockets, court documents, and report search.
- pgvector for lightweight semantic search if OpenSearch vector is not needed immediately.

7.4 Why this minimizes cost

- Object storage is cheaper than warehouses for raw data.
- DuckDB can query Parquet without running a warehouse cluster.
- Postgres remains simple and cheap for operational workloads.
- OpenSearch is used only for document search, not as the system of record.
- Graph is implemented in Postgres first to avoid early graph database complexity.

7.5 Best for

- First product release.
- Small to mid-sized enterprise pilot.
- Limited engineering team.
- Sources with moderate ingestion volume.
- Reports and dashboards that do not require sub-second analytics over billions of rows.

7.6 Risks

- Postgres graph queries may become slow for deep pathfinding.
- OpenSearch can become expensive if all raw documents are indexed indiscriminately.
- DuckDB is excellent for batch/local analytics but not a high-concurrency warehouse.

7.7 Mitigations

- Keep graph depth defaults small.
- Store raw documents in object storage and index only extracted text/chunks that are useful.
- Partition Parquet by source/date/entity type.
- Add ClickHouse or Iceberg later for scale.

7.8 Proposal B - Lakehouse Scale-Out

7.9 Recommended when raw data volume grows

Use:

- Object storage: S3, Cloudflare R2, GCS, Azure Blob, or MinIO.
- Open table format: Apache Iceberg.
- File format: Parquet.
- Query engine: Trino, Athena, Spark, or DuckDB for smaller jobs.
- Catalog: AWS Glue, Nessie, Hive Metastore, or managed catalog depending on cloud.

7.10 Why

Apache Iceberg is designed for huge analytic tables and adds table-like reliability to data lakes, including schema evolution, snapshots, and multi-engine access. This is useful when the platform has large source-history tables, market data, filings, comments, contracts, and derived relationship events.

7.11 Best for

- Multi-terabyte public-record archive.
- Historical filings and public comments.
- Time-travel data reproducibility.
- Cost-efficient long-term raw/curated data retention.
- Multiple analytics engines.

7.12 Risks

- More infrastructure complexity.
- Requires data engineering maturity.
- Small-file management and compaction jobs must be maintained.

7.13 Mitigations

- Start with Parquet layout that can migrate to Iceberg.
- Add Iceberg only when datasets and team maturity justify it.
- Build compaction and partition hygiene jobs early.

7.14 Proposal C - Managed Warehouse

7.15 Alternatives

- BigQuery.
- Snowflake.
- Databricks SQL.
- MotherDuck for smaller DuckDB-compatible workloads.

7.16 Best for

- Enterprise customers that already use a warehouse.
- BI dashboards with many concurrent users.
- Business users who want SQL access.
- Fast ad-hoc analytics without managing query clusters.
- Data sharing with enterprise customers.

7.17 Cost considerations

- Warehouses can become expensive if raw API responses, PDF text, embeddings, and high-cardinality event data are loaded indiscriminately.
- Prefer warehouse for curated tables, not the raw document vault.
- Keep raw evidence in object storage.
- Export curated facts, entities, relationships, financial metrics, and daily aggregates to the warehouse.

7.18 Recommended usage

Use managed warehouse as an optional enterprise add-on or internal analytics tier after product-market fit.

7.19 Proposal D - Real-Time Analytics with ClickHouse

7.20 Use ClickHouse when

- Alert/event volume becomes high.
- Users need fast aggregations over relationship events, source runs, price bars, news signals, and filings.
- Dashboards need real-time analytics at low latency.
- Cost of managed warehouse queries becomes too high.

7.21 Good ClickHouse tables

- market_bars.
- source_events.
- relationship_events.
- alert_events.
- document_mentions.
- entity_metrics_daily.
- portfolio_metrics_daily.

7.22 Do not use ClickHouse for

- Primary transactional user data.
- Report editor state.
- Entity merge transactions.
- Permission checks.

- Raw document storage.

7.23 API-First vs Ingest-First Strategy

7.24 API-first/on-demand

Use for:

- Low-frequency sources.
- Expensive licensed data.
- Data with restrictive licensing.
- Very large sources where most records are never queried.

Pros:

- Lower storage cost.
- Faster to implement.
- Less compliance burden.

Cons:

- Slower reports.
- External API dependency.
- Rate-limit risk.
- Harder reproducibility.

7.25 Ingest-first/cache locally

Use for:

- High-value public records.
- Sources needed for graph/pathfinding.
- Sources needed for alerts.
- Sources needed for reproducible reports.

Pros:

- Faster user experience.
- Better graph analytics.
- Better evidence preservation.
- Lower per-query external API cost.

Cons:

- More storage.
- More maintenance.
- Compliance and retention policies required.

7.26 Recommended hybrid

- Phase 1: fully ingest high-value U.S. public records and cache market/financial data selectively.
- Phase 2: use API-first for new global sources, then promote high-value sources to scheduled ingestion.

7.27 Cost Minimization Rules

- Store raw evidence once, compressed, content-addressed by hash.
- Deduplicate identical documents and API responses.
- Partition object storage by source/year/month/day and record type.
- Index only extracted text, metadata, and useful chunks, not every raw byte.
- Use hot/warm/cold storage tiers.
- Use incremental syncs and checkpoints instead of full refreshes.
- Cache expensive API calls with source-specific TTLs.
- Materialize only frequently used graph edges and report-ready facts.
- Use DuckDB/Parquet before paying for a warehouse.
- Use ClickHouse for high-volume analytics when Postgres/OpenSearch become costly.
- Do not store licensed market data longer than allowed by vendor terms.
- Let enterprise customers bring their own market-data credentials where possible.
- Add cost attribution by tenant, source, connector, report, and model provider.

8 Core Data Model and API Contracts

8.1 Canonical Entity Schema

```
{
  "id": "entity_uuid",
  "tenant_id": "uuid",
  "type": "Company | Person | Fund | Agency | Bill | Regulation | Docket | ContractAward |
CourtCase | Nonprofit | PAC | Security | Document",
  "canonical_name": "Example Corp",
  "aliases": ["Example Corporation"],
  "identifiers": [
    {"scheme": "CIK", "value": "0000000000", "source": "SEC"},
    {"scheme": "LEI", "value": "...", "source": "GLEIF"}
  ],
  "jurisdictions": ["US-DE", "US"],
  "description": "short generated or analyst-authored description",
  "confidence": 0.95,
  "visibility": "workspace | organization | private",
  "created_at": "timestamp",
  "updated_at": "timestamp"
}
```

8.2 Relationship Schema

```
{
  "id": "relationship_uuid",
  "tenant_id": "uuid",
  "subject_entity_id": "uuid",
  "predicate": "LOBBIED_ON",
  "object_entity_id": "uuid",
  "start_date": "2025-01-01",
  "end_date": "2025-03-31",
  "amount": 120000,
  "currency": "USD",
  "confidence": 0.91,
  "evidence_refs": ["evidence_uuid"],
  "source_record_ids": ["source_record_uuid"],
  "created_at": "timestamp",
  "updated_at": "timestamp"
}
```

8.3 Claim Schema

```
{
  "id": "claim_uuid",
  "report_id": "uuid",
  "claim_text": "Company X disclosed lobbying on defense procurement issues in Q2 2025.",
  "claim_type": "lobbying | financial | valuation | regulatory | procurement | legal | ownership | risk",
  "confidence_label": "confirmed | likely | possible | unverified | contradicted",
  "evidence_refs": ["evidence_uuid"],
  "contradictions": [],
  "review_status": "draft | needs_review | approved | rejected",
  "created_by": "user_uuid",
  "reviewed_by": "user_uuid",
  "created_at": "timestamp",
  "updated_at": "timestamp"
}
```

8.4 API Endpoint Map

8.5 Search and entities

```
GET /api/search?q=&types=&sources=&date_from=&date_to=&confidence_min=
GET /api/entities/{id}
GET /api/entities/{id}/timeline
GET /api/entities/{id}/relationships
```

GET /api/entities/{id}/evidence
POST /api/entities/{id}/related-parties
POST /api/entities/merge-candidates
POST /api/entities/{id}/merge
POST /api/entities/{id}/unmerge

8.6 Graph

POST /api/graph/path
POST /api/graph/expand
POST /api/graph/query
POST /api/graph/export

8.7 Sources

GET /api/sources
GET /api/sources/{id}
POST /api/sources/{id}/run
GET /api/source-runs/{id}
GET /api/source-runs/{id}/errors
GET /api/source-records/{id}

8.8 Finance

POST /api/finance/analyze-stock
POST /api/finance/valuation/dcf
POST /api/finance/comps
POST /api/finance/technicals
POST /api/finance/earnings-summary
POST /api/finance/portfolio/analyze

8.9 Claude finance skills gateway

GET /api/skills
POST /api/skills/run
GET /api/skills/runs/{id}
POST /api/skills/runs/{id}/approve
POST /api/skills/runs/{id}/reject

8.10 Reports

POST /api/reports
GET /api/reports/{id}
POST /api/reports/{id}/generate
POST /api/reports/{id}/sections/{section_id}/enhance
POST /api/reports/{id}/claims/{claim_id}/approve
POST /api/reports/{id}/claims/{claim_id}/reject
POST /api/reports/{id}/export

GET /api/reports/{id}/versions
POST /api/reports/{id}/compare

8.11 Alerts and watchlists

POST /api/watchlists
POST /api/watchlists/{id}/items
POST /api/alerts
GET /api/alerts/events
POST /api/alerts/{id}/pause
POST /api/alerts/{id}/resume

9 UX Requirements

9.1 Home Dashboard

- Recent alerts
- Watched entities
- Portfolio risks
- New SEC filings
- New lobbying reports
- New contracts
- New regulatory changes
- Reports awaiting review
- Connector health

9.2 Search Page

- Global search
- Entity filters
- Source filters
- Date filters
- Saved searches
- Recent searches
- Confidence filters

9.3 Entity Profile

- Overview
- Graph
- Timeline
- Evidence
- Filings
- Financials
- Lobbying
- Political money

- Regulations
- Contracts
- Courts
- Sanctions
- Related parties
- Reports
- Notes

9.4 Stock Analysis Page

- Snapshot
- Chart
- Fundamentals
- Valuation
- Technicals
- Earnings
- Ownership
- Catalysts
- Risks
- Government exposure
- Reports

9.5 Investigation Workspace

- Case board
- Entity graph
- Evidence board
- Timeline
- Notes
- Tasks
- Claims
- Report draft

9.6 Report Editor

- Template selector
- Editable sections
- Evidence sidebar
- Claim cards
- Confidence labels
- Citation verifier
- Comments
- Suggested changes
- Version history
- Export buttons

9.7 Admin Dashboard

- Users
- Roles
- Workspaces
- Sources
- Connectors
- API keys
- Audit logs
- Usage
- Billing hooks
- Security settings
- Model settings

10 Development Plan

10.1 Codex-Based Development Method

Use Codex as a development accelerator with human architecture and code review.

10.2 Workflow

1. Human architect defines epics and acceptance criteria.
2. Codex implements scoped PRs.
3. Codex writes tests and fixtures.
4. CI runs lint, typing, unit tests, integration test stubs, and security checks.
5. Codex fixes CI failures if assigned.
6. Human reviewer checks architecture, security, source usage, and business logic.
7. Human merges.

10.3 AGENTS.md rules

- Use official APIs and bulk data before scraping.
- Never bypass CAPTCHA, authentication, paywalls, robots restrictions, or rate limits.
- Every connector must store raw records before normalization.
- Every normalized entity or relationship must include EvidenceRef.
- Every connector must include tests and fixtures.
- Every source must include source_contract.yml.
- Every financial conclusion must expose assumptions and risk factors.
- Do not generate unsupported allegations.
- Do not merge entities without deterministic identifiers or approved confidence.
- Do not expose sensitive SAM.gov data.
- Treat FEC individual contributor data as restricted-use.
- Generated reports require citation verification.

10.4 Phase 1 Development Timeline

10.4.1 P1.0 Foundation (2-4 weeks)

- Create monorepo
- Docker compose
- FastAPI skeleton
- Next.js skeleton
- Postgres migrations
- object storage abstraction
- auth skeleton
- audit log skeleton
- CI/CD
- AGENTS.md
- source contract template

10.4.2 P1.1 Enterprise Core (4-6 weeks)

- Organizations
- workspaces
- users
- roles
- permissions
- projects/cases
- watchlists
- admin dashboard
- audit log UI
- source registry
- evidence vault MVP

10.4.3 P1.2 U.S. Connector Wave 1 (6-8 weeks)

- SEC EDGAR
- FEC
- LDA.gov
- Congress.gov
- Federal Register
- USAspending
- OpenCorporates
- GLEIF
- OpenSanctions/OFAC

10.4.4 P1.3 Entity Graph (6-8 weeks)

- canonical ontology
- identifier model
- relationship model

- deterministic matching
- fuzzy matching
- merge queue
- graph API
- graph UI
- timeline UI
- related-party discovery MVP

10.4.5 P1.4 Investor Intelligence (6-10 weeks)

- market data provider abstraction
- financial statements
- ratios
- valuation
- DCF
- comps
- technical analysis
- insider/institutional ownership
- stock analysis dashboard

10.4.6 P1.5 Claude Finance Skills and Reports (6-10 weeks)

- FinanceSkillGateway
- Anthropic financial-services adapter
- comps
- DCF
- earnings
- one-pager
- IC memo
- evidence bundles
- claim generator
- citation verifier
- report editor
- review workflow

10.4.7 P1.6 Connector Wave 2 and Monitoring (6-8 weeks)

- FARA
- GovInfo
- Regulations.gov
- eCFR
- RegInfo/OIRA
- SAM.gov
- IRS 990
- CourtListener/RECAP

- alerts
- portfolio import

10.4.8 P1.7 Enterprise Hardening (6-10 weeks)

- SSO/SAML/OIDC
- SCIM-ready model
- field-level permissions
- restricted source controls
- export approvals
- load tests
- security review
- tenant isolation tests
- backup/restore
- observability

10.5 Phase 2 Development Timeline

10.5.1 P2.0 Global Framework (4-6 weeks)

- jurisdiction model
- multilingual aliases
- country-specific identifiers
- currency handling
- translation pipeline
- global source contracts

10.5.2 P2.1 U.K. and EU (8-12 weeks)

- Companies House
- UK PSC
- EU Transparency Register
- EU TED
- EU FTS
- EU sanctions
- U.K. procurement and political finance sources

10.5.3 P2.2 Canada and Australia (6-10 weeks)

- Canada lobbying
- Elections Canada
- CanadaBuys
- AEC Transparency Register
- Australia lobbyists
- AusTender
- ASIC

10.5.4 P2.3 Brazil and Latin America seed (8-12 weeks)

- Brazil Portal da Transparencia
- Compras.gov.br
- Camara
- Senado
- CVM
- TSE
- e-Agendas

10.5.5 P2.4 Cross-Border Investigative Sources (8-12 weeks)

- ICIJ Offshore Leaks
- OCCRP/OpenAleph
- OpenOwnership/BODS
- World Bank procurement
- UNGM
- global sanctions/debarment

10.5.6 P2.5 Global Reports and Advanced Analytics (ongoing)

- cross-border ownership report
- global sanctions/PEP report
- international procurement exposure
- global regulatory exposure
- portfolio geopolitical risk report

11 Detailed Engineering Backlog

11.1 Epic 1 - Platform Foundation

- Create monorepo
- Configure Docker
- Configure CI/CD
- FastAPI app
- Next.js app
- Postgres migrations
- object storage client
- Redis
- logging
- error monitoring
- test framework
- AGENTS.md

11.2 Epic 2 - Enterprise Auth and Governance

- User model

- Organization model
- Workspace model
- Roles
- Permissions
- Invite flow
- SSO hooks
- SCIM-ready schema
- MFA hooks
- Audit logs

11.3 Epic 3 - Source Registry and Evidence Vault

- Source model
- Source credentials
- Source contract
- Source run
- Raw document storage
- Document hashing
- EvidenceRef model
- Evidence viewer
- Source health dashboard

11.4 Epic 4 - U.S. Connectors

- SEC
- FEC
- LDA
- FARA
- Congress
- GovInfo
- Federal Register
- Regulations.gov
- eCFR
- RegInfo/OIRA
- USAspending
- SAM.gov
- IRS 990
- CourtListener
- OpenSanctions/OFAC
- OpenCorporates
- GLEIF

11.5 Epic 5 - Ontology and Entity Resolution

- Entity model

- Identifier model
- Relationship model
- deterministic matching
- Splink/fuzzy matching
- merge queue
- unmerge
- block merge
- conflict detection
- entity history

11.6 Epic 6 - Graph Intelligence

- Graph API
- pathfinding
- related-party discovery
- graph UI
- timeline UI
- edge evidence viewer
- graph export

11.7 Epic 7 - Financial Intelligence

- market data providers
- fundamentals
- ratios
- DCF
- comps
- technical indicators
- earnings summary
- insider activity
- institutional holdings
- portfolio analytics

11.8 Epic 8 - Claude Finance Skills Gateway

- skill registry
- Anthropic adapter
- comps skill
- DCF skill
- earnings skill
- one-pager skill
- IC memo skill
- due diligence skill
- sandbox
- audit logging

- review integration

11.9 Epic 9 - Report Engine

- report templates
- evidence bundle builder
- claim builder
- citation verifier
- contradiction detector
- report editor
- versioning
- comments
- redlines/suggestions
- exports

11.10 Epic 10 - Alerts and Portfolio

- portfolio model
- CSV import
- watchlists
- alert rules
- alert event scanner
- email
- Slack
- Teams
- webhooks
- daily/weekly briefs

11.11 Epic 11 - Admin and Compliance

- source policy registry
- FEC restrictions
- SAM public/sensitive separation
- export approvals
- data retention
- legal hold
- usage analytics
- billing hooks

12 Testing, Quality, and Release Requirements

12.1 Test types

- Unit tests for parsers, normalizers, validators, and financial calculations.
- Contract tests for every connector against saved fixtures.

- Integration tests against live APIs behind explicit environment flags.
- Golden-file tests for source parsing.
- Entity-resolution regression tests.
- Report citation-verifier tests.
- Security tests for permissions and tenant isolation.
- Load tests for search, graph expansion, and report generation.
- Data-quality tests for freshness, duplicates, missing IDs, and parse failures.

12.2 CI gates

- Lint passes.
- Type checking passes.
- Unit tests pass.
- Contract tests pass.
- No high-severity dependency vulnerabilities.
- Migrations are reversible or explicitly approved.
- Source contracts updated when connector behavior changes.
- Report templates pass citation-verifier tests.

12.3 Release environments

- Local development.
- Shared development.
- Staging with test credentials.
- Production.
- Optional customer-dedicated environment for enterprise clients.

13 Security, Compliance, and Governance

13.1 Security requirements

- Tenant isolation.
- Encryption in transit.
- Encryption at rest.
- Secrets manager.
- Role-based access control.
- Attribute-based access control.
- SSO/SAML/OIDC.
- MFA.
- Audit logs.
- IP allowlisting for enterprise.
- Secure export links.
- Dependency scanning.
- Container scanning.

- Backups.
- Disaster recovery.

13.2 Compliance controls

- FEC individual contributor data must be restricted from solicitation or prohibited commercial misuse.
- SAM.gov sensitive API fields must not be displayed or disseminated outside permitted use.
- Reports must avoid unsupported allegations.
- Financial reports must expose assumptions and risks.
- High-risk claims require review.
- Private residential addresses should be masked unless explicitly approved and legally necessary.
- Source licenses and terms must be tracked.
- Market-data vendor retention rules must be enforced.

14 Feature Parameter Reference

This section defines implementation-level parameters for the highest-value user actions. These should become the starting point for API schemas, frontend forms, validation rules, permissions, and test fixtures.

14.1 Generate Intelligence Report

```
{
  "report_type": "stock_analysis|company_intelligence|long_thesis|short_thesis|
investment_committee_memo|lobbying_influence|regulatory_exposure|
government_contracting|litigation_risk|related_party|portfolio_risk|
vendor_due_diligence",
  "primary_entity_id": "uuid",
  "secondary_entity_ids": ["optional related entities"],
  "scope": {
    "jurisdictions": ["US"],
    "date_from": "2018-01-01",
    "date_to": "2026-05-25",
    "sources": ["SEC", "FEC", "LDA", "USAspending", "FederalRegister", "CourtListener"],
    "include_possible_matches": false,
    "min_confidence": 0.70
  },
  "finance": {
    "include_dcf": true,
    "include_comps": true,
    "include_technicals": true,
    "include_earnings": true,
  }
}
```



```

    "use_claude_finance_skills": true,
    "peer_group": [],
    "time_horizon": "12-36_months"
  },
  "report_style": {
    "audience": "investment_committee|executive|analyst|compliance|trader",
    "depth": "brief|standard|institutional|deep_dive",
    "tone": "neutral|investor_memo|board_brief|technical|compliance",
    "include_appendix": true,
    "include_evidence_table": true,
    "include_graph_images": true,
    "include_financial_model": true
  },
  "review": {
    "review_required": true,
    "reviewers": ["user_uuid"],
    "block_uncited_claims": true,
    "block_high_risk_claims_without_approval": true
  }
}

```

14.2 Enhance Report Section

```

{
  "report_id": "uuid",
  "section_id": "uuid",
  "selected_text_range": {"start": 120, "end": 980},
  "operation": "expand|condense|rewrite|add_citations|add_bear_case|add_bull_case|
add_risks|add_financial_detail|run_dcf|run_comps|run_earnings_update|
make_executive_ready|make_compliance_safe",
  "allowed_sources": ["current_evidence_bundle", "new_retrieval",
"user_selected_sources"],
  "style": "institutional|plain_english|technical|board_memo|trader_brief",
  "change_mode": "suggestion|direct_edit|new_version",
  "requires_review": true
}

```

14.3 Query Graph Path

```

{
  "from_entity_id": "uuid",
  "to_entity_id": "uuid",
  "max_depth": 4,
  "relationship_types": ["LOBBIED_FOR", "LOBBIED_ON", "RECEIVED_CONTRACT",
"OFFICER_OF", "DONATED_TO", "SUBMITTED_COMMENT"],
}

```

```
"date_from": "2016-01-01",
"date_to": "2026-05-25",
"min_confidence": 0.70,
"max_paths": 25,
"ranking": "shortest|highest_confidence|largest_amount|most_recent|
investor_relevance"
}
```

14.4 Run Source Connector

```
{
  "source": "sec_edgar",
  "mode": "incremental|backfill|single_entity|repair|dry_run",
  "entity_ids": ["optional uuid"],
  "source_native_ids": ["CIK or other source ID"],
  "date_from": "2024-01-01",
  "date_to": "2026-05-25",
  "store_raw": true,
  "normalize": true,
  "create_relationships": true,
  "index_documents": true,
  "respect_rate_limits": true
}
```

14.5 Portfolio Import

```
{
  "portfolio_name": "Core Long Book",
  "base_currency": "USD",
  "positions": [
    {"ticker": "PLTR", "quantity": 1000, "cost_basis": 10.50, "as_of_date": "2026-05-25"}
  ],
  "mapping_mode": "auto_with_review",
  "analysis_modules": ["valuation", "risk", "regulatory", "contracts", "earnings",
"technicals"],
  "alert_defaults": {
    "sec_filings": true,
    "earnings": true,
    "contracts": true,
    "lobbying": true,
    "regulations": true,
    "price_moves": true
  }
}
```

15 Detailed Report Templates

15.1 Stock Analysis Report Template

Required sections:

1. Report scope and data cutoff.
2. Executive summary.
3. Company overview.
4. Stock snapshot.
5. Price action and liquidity.
6. Fundamental analysis.
7. Financial statement trends.
8. Valuation summary.
9. DCF model and assumptions.
10. Comparable company analysis.
11. Technical analysis.
12. Earnings and guidance.
13. Insider and institutional ownership.
14. Government-contract exposure.
15. Lobbying and regulatory exposure.
16. Litigation, enforcement, sanctions, and exclusions.
17. Related parties.
18. Bull case.
19. Base case.
20. Bear case.
21. Catalysts.
22. Risks.
23. Monitoring plan.
24. Evidence appendix.

15.2 Company Intelligence Report Template

Required sections:

1. Scope and methodology.
2. Executive summary.
3. Corporate identity and identifiers.
4. Ownership and corporate structure.
5. Executives, board, officers, and advisers.
6. Subsidiaries and related companies.
7. Investors and funds.
8. SEC/public filings.

9. Government contracts and grants.
10. Lobbying and political-money footprint.
11. Regulatory exposure.
12. Litigation and enforcement.
13. Sanctions and exclusions.
14. Nonprofit/foundation/trade-association links.
15. Related-party map.
16. Timeline.
17. Key findings.
18. Open questions.
19. Evidence appendix.

15.3 Lobbying Influence Report Template

Required sections:

1. Scope.
2. Executive summary.
3. Lobbying clients and registrants.
4. Lobbying firms and lobbyists.
5. Issues lobbied.
6. Agencies lobbied.
7. Bills referenced.
8. Quarterly amounts.
9. Public comments and OIRA meetings.
10. Related contractors and procurement exposure.
11. Related PACs and political-money context where allowed.
12. Revolving-door indicators.
13. Timeline.
14. Confidence and limitations.
15. Evidence appendix.

15.4 Government Contracting Report Template

Required sections:

1. Scope.
2. Executive summary.
3. Award summary by agency.
4. Award summary by year.
5. Contract types.
6. NAICS/PSC exposure.
7. Subawards and partners.
8. Concentration risks.

9. SAM opportunities and exclusions.
10. Lobbying overlap.
11. Regulatory/bill overlap.
12. Litigation and enforcement.
13. Procurement red flags.
14. Investor implications.
15. Evidence appendix.

15.5 Portfolio Risk Report Template

Required sections:

1. Portfolio overview.
2. Allocation and concentration.
3. Performance summary.
4. Valuation summary.
5. Earnings calendar.
6. Catalyst calendar.
7. Regulatory exposure.
8. Government-contract exposure.
9. Lobbying exposure.
10. Litigation and sanctions risk.
11. Technical risk and volatility.
12. Position-level thesis tracker.
13. Highest priority research actions.
14. Alerts triggered.
15. Evidence appendix.

16 Database Schema - Implementation Detail

16.1 Operational Postgres tables

organizations

workspaces

users

roles

permissions

workspace_memberships

projects

cases

watchlists

watchlist_items

portfolios

portfolio_positions
sources
source_credentials
source_contracts
source_runs
source_records
raw_documents
evidence_refs
entities
entity_aliases
entity_identifiers
entity_descriptions
relationships
relationship_evidence
entity_merge_candidates
entity_merge_events
claims
claim_evidence
reports
report_sections
report_versions
report_comments
report_suggestions
report_exports
skills
skill_runs
skill_artifacts
alerts
alert_rules
alert_events
audit_logs
data_quality_issues

16.2 Recommended indexes

entities(type, canonical_name)
entity_identifiers(scheme, value)
relationships(subject_entity_id, predicate)
relationships(object_entity_id, predicate)
relationships(predicate, start_date)
source_records(source_id, source_native_id)
source_records(source_id, published_at)
evidence_refs(source_record_id)
claims(report_id, review_status)

alerts(workspace_id, enabled)
portfolio_positions(portfolio_id, ticker)
audit_logs(workspace_id, created_at)

16.3 Partition candidates

- source_records by source and month when high volume.
- raw_documents metadata by source and month; body stays in object storage.
- relationships by predicate or time if extremely large.
- alert_events by month.
- audit_logs by month.
- Market bars by ticker/date in analytical storage, not operational Postgres.

17 Source Refresh Cadence Recommendations

SEC EDGAR: hourly for watched companies; daily broad sync; historical backfill by batch.

FEC: daily/nightly sync; report-specific backfill on demand.

LDA.gov: daily; quarterly filing windows get higher frequency.

FARA: daily if bulk updated daily; otherwise scheduled daily/weekly.

Congress.gov: daily for watched bills; weekly broad sync.

Federal Register: daily morning sync; alert watched agencies/rules.

Regulations.gov: daily for watched dockets; backfill comments in batches.

eCFR: daily recent changes; weekly full title snapshots for watched titles.

RegInfo/OIRA: daily watched RINs; weekly broad agenda/review sync.

USAspending: daily for watched recipients/agencies; weekly broad sync.

SAM.gov: daily watched opportunities and exclusions.

IRS 990: monthly or quarterly depending on IRS update cadence.

CourtListener: daily watched entities/cases; weekly broad legal-risk search.

OpenSanctions/OFAC: daily.

OpenCorporates/GLEIF: on-demand plus weekly refresh for watched entities.

Market data: real-time or delayed depending on license; daily fundamentals refresh.

18 Data Quality and Observability

18.1 Data quality checks

- Source run completed successfully.
- Expected record count within historical range.
- API schema unchanged or migration handled.
- Required identifiers present.
- Date fields parse correctly.
- Amount/currency fields parse correctly.
- Duplicates below threshold.

- Broken document URLs below threshold.
- Raw document hash stored.
- Normalized entities created.
- Evidence references attached.
- Graph edges have confidence scores.
- No high-risk report claim without evidence.

18.2 Connector metrics

- Records discovered.
- Records fetched.
- Records skipped.
- Records parsed.
- Records failed.
- Raw bytes stored.
- Entities created.
- Relationships created.
- EvidenceRefs created.
- API requests.
- Rate-limit events.
- Retry count.
- Cost estimate.

18.3 User-facing freshness display

Every entity profile and report should show:

- Last source refresh.
- Source coverage.
- Sources included.
- Sources excluded.
- Data cutoff.
- Known gaps.

19 Architecture Decision Matrix

19.1 Storage and analytics options

Option A: Postgres + object storage + DuckDB + OpenSearch

Best for: Phase 1 MVP, low cost, fast build.

Pros: simple, cheap, developer-friendly.

Cons: limited high-concurrency analytics.

Decision: recommended starting architecture.

Option B: Iceberg lakehouse + object storage + Trino/Athena/Spark

Best for: large historical public-record archive.

Pros: scalable, open format, low-cost storage, schema evolution.

Cons: more data engineering complexity.

Decision: prepare Parquet layout now; adopt in Phase 2 or high-scale Phase 1.

Option C: BigQuery/Snowflake/Databricks

Best for: enterprise BI and SQL workloads.

Pros: managed, scalable, strong integrations.

Cons: can get expensive; raw document storage not ideal.

Decision: optional enterprise add-on; curated tables only.

Option D: ClickHouse

Best for: real-time analytics, event streams, dashboards, alerts.

Pros: fast OLAP and cost-effective compression.

Cons: not an OLTP store; operational mutations are not its strength.

Decision: add when alert/event analytics need it.

19.2 Graph database options

Option A: Postgres edge tables

Best for: Phase 1, controlled graph depth, low complexity.

Decision: start here.

Option B: Neo4j

Best for: analyst graph exploration and mature graph algorithms.

Decision: evaluate after graph query patterns are proven.

Option C: ArangoDB

Best for: multi-model document/graph use cases.

Decision: consider if document/graph integration becomes central.

Option D: Amazon Neptune

Best for: AWS enterprise managed graph deployments.

Decision: consider for enterprise AWS-only deployments.

19.3 Search/vector options

Option A: Postgres full-text + pgvector

Best for: low-cost MVP, smaller corpus.

Decision: use for early semantic prototypes.

Option B: OpenSearch

Best for: scalable full-text search, filtering, vector search, document corpus.

Decision: recommended Phase 1 search layer.

Option C: dedicated vector DB such as Qdrant, Weaviate, Pinecone

Best for: large-scale RAG and embedding-heavy workloads.
Decision: postpone until OpenSearch/pgvector limits are proven.

20 Developer Launch Checklist

Before coding starts:

- Create GitHub repository.
- Add AGENTS.md.
- Add README with product scope.
- Add source contract template.
- Add Docker Compose.
- Add FastAPI skeleton.
- Add Next.js skeleton.
- Add Postgres migrations.
- Add object storage abstraction.
- Add CI with lint, tests, typing.
- Add issue templates.
- Add PR template.
- Add coding standards.
- Create epics and labels.
- Create source connector tickets.
- Create report template tickets.
- Create architecture decision records.
- Assign humans to security/compliance review.

Before Phase 1 launch:

- Core U.S. source connectors pass tests.
- Every fact in reports has evidence.
- Entity merges are reversible.
- Report editor supports review and enhancement.
- Claude financial skills are integrated through controlled gateway.
- FEC and SAM restrictions are enforced.
- Exports are permissioned and audited.
- Tenant isolation is tested.
- Cost dashboard is available for sources, storage, search, and model usage.

21 Appendix A - Organized URLs

The following full URLs should be used by developers for source contracts, integration planning, API-client implementation, and documentation. Verify terms, rate limits, and authentication requirements before production use.

21.1 Official U.S. public-record APIs and data sources

- **SEC EDGAR APIs:** <https://www.sec.gov/search-filings/edgar-application-programming-interfaces>
- **SEC data APIs base:** <https://data.sec.gov/>
- **SEC developer resources and fair access:** <https://www.sec.gov/about/developer-resources>
- **SEC accessing EDGAR data:** <https://www.sec.gov/search-filings/edgar-search-assistance/accessing-edgar-data>
- **FEC OpenFEC API docs:** <https://api.open.fec.gov/developers/>
- **FEC data portal:** <https://www.fec.gov/data/>
- **FEC contributor data usage restrictions:** <https://www.fec.gov/updates/sale-or-use-contributor-information/>
- **LDA.gov API documentation:** <https://lda.gov/api/redoc/v1/>
- **LDA.gov API landing page:** <https://lda.gov/api/>
- **Senate LDA API landing page:** <https://lda.senate.gov/api/>
- **FARA bulk data and API:** <https://efile.fara.gov/ords/fara/f?p=1235:10>
- **DOJ FARA:** <https://www.justice.gov/nsd-fara>
- **Congress.gov API:** <https://api.congress.gov/>
- **Library of Congress Congress.gov API info:** <https://www.loc.gov/apis/additional-apis/congress-dot-gov-api/>
- **GovInfo API docs:** <https://api.govinfo.gov/docs/>
- **GovInfo API app page:** <https://www.govinfo.gov/app/api>
- **Federal Register API docs:** <https://www.federalregister.gov/developers/documentation/api/v1>
- **Federal Register site:** <https://www.federalregister.gov/>
- **Regulations.gov API docs:** <https://open.gsa.gov/api/regulationsgov/>
- **Regulations.gov:** <https://www.regulations.gov/>

- **eCFR API docs:** <https://www.ecfr.gov/developers/documentation/api/v1>
- **eCFR:** <https://www.ecfr.gov/>
- **RegInfo:** <https://www.reginfo.gov/public/>
- **Unified Agenda main page:** <https://www.reginfo.gov/public/do/eAgendaMain>
- **OIRA review search:** <https://www.reginfo.gov/public/do/eoReviewSearch>
- **GSA RegInfo overview:** <https://www.gsa.gov/policy-regulations/policy/federal-regulation-policy/regulatory-information-reginfogov>
- **USAspending API:** <https://api.usaspending.gov/>
- **USAspending API endpoints:** <https://api.usaspending.gov/docs/endpoints>
- **USAspending:** <https://www.usaspending.gov/>
- **SAM.gov:** <https://sam.gov/>
- **GSA Open APIs index:** <https://open.gsa.gov/api/>
- **SAM opportunities API:** <https://open.gsa.gov/api/get-opportunities-public-api/>
- **SAM entity extracts API:** <https://open.gsa.gov/api/sam-entity-extracts-api/>
- **SAM exclusions API:** <https://open.gsa.gov/api/exclusions-api/>
- **SAM terms of use:** <https://sam.gov/about/terms-of-use>
- **IRS Form 990 downloads:** <https://www.irs.gov/charities-non-profits/form-990-series-downloads>
- **IRS tax-exempt bulk data:** <https://www.irs.gov/charities-non-profits/tax-exempt-organization-search-bulk-data-downloads>
- **CourtListener:** <https://www.courtlistener.com/>
- **CourtListener RECAP:** <https://www.courtlistener.com/recap/>
- **CourtListener API docs:**
<https://wiki.free.law/c/courtlistener/help/api/rest/v4/overview>
- **OFAC:** <https://ofac.treasury.gov/>
- **OFAC sanctions search:** <https://sanctionssearch.ofac.treas.gov/>
- **OFAC SDN XML:** <https://www.treasury.gov/ofac/downloads/sdn.xml>
- **OFAC consolidated XML:**
<https://www.treasury.gov/ofac/downloads/consolidated/consolidated.xml>

- **OpenCorporates API docs:** <https://api.opencorporates.com/documentation/API-Reference>
- **OpenCorporates:** <https://opencorporates.com/>
- **GLEIF API:** <https://www.gleif.org/en/lei-data/gleif-api>
- **GLEIF concatenated files:** <https://www.gleif.org/en/lei-data/gleif-concatenated-file>

21.2 U.S. GitHub repos and open-source connector tools

- **SEC edgartools:** <https://github.com/dgunning/edgartools>
- **SEC EDGAR downloader:** <https://github.com/jadchaar/sec-edgar-downloader>
- **sec-edgar CLI:** <https://github.com/sec-edgar/sec-edgar>
- **sec-api Python client:** <https://github.com/janlukasschroeder/sec-api-python>
- **sec-api Node client:** <https://github.com/janlukasschroeder/sec-api-node>
- **SEC EDGAR toolkit:** <https://github.com/stefanoamorelli/sec-edgar-toolkit>
- **SEC EDGAR MCP:** <https://github.com/stefanoamorelli/sec-edgar-mcp>
- **sec-edgar-api docs:** <https://sec-edgar-api.readthedocs.io/>
- **FEC openFEC:** <https://github.com/fecgov/openFEC>
- **libfec:** <https://github.com/asg017/libfec>
- **fecfile parser:** <https://github.com/esonderegger/fecfile>
- **NYTimes Fech:** <https://github.com/NYTimes/Fech>
- **R.openFEC:** <https://github.com/robinspollak/R.openFEC>
- **fecapi R:** <https://github.com/jonthegeek/fecapi>
- **fec-data Postgres import:** <https://github.com/lbeaufort/fec-data>
- **lobbyr:** <https://github.com/Lobbying-Disclosure/lobbyr>
- **Politico Senate lobbying scraper:** https://github.com/The-Politico/scraper_senate-lobbying-disclosures
- **LobbyView Python package:**
<https://github.com/lobbyview/LobbyViewPythonPackage>
- **FARA Datasette:** <https://github.com/simonw/fara-datasette>
- **FARA history:** <https://github.com/simonw/fara-history>
- **Sunlight FARA:** <https://github.com/sunlightlabs/fara>

- **LibraryOfCongress Congress.gov API repo:**
<https://github.com/LibraryOfCongress/api.congress.gov>
- **unitedstates/congress:** <https://github.com/unitedstates/congress>
- **unitedstates/congress-legislators:** <https://github.com/unitedstates/congress-legislators>
- **R congress package:** <https://github.com/christopherkenny/congress>
- **US Congress Python API:** <https://github.com/areed1192/united-states-congress-python-api>
- **GPO API repo:** <https://github.com/usgpo/api>
- **GPO link service:** <https://github.com/usgpo/link-service>
- **DataMade GovInfo:** <https://github.com/datamade/govinfo>
- **GovInfo PHP:** <https://github.com/cnizzardini/gov-info>
- **Federal Register API core:** <https://github.com/usnationalarchives/federalregister-api-core>
- **Federal Register web:** <https://github.com/usnationalarchives/federalregister-web>
- **Federal Register Ruby client:**
https://github.com/usnationalarchives/federal_register
- **R Federal Register:** <https://github.com/ropengov/federalregister>
- **US Federal Register Python API:** <https://github.com/areed1192/us-federal-register-python-api>
- **Regulations public comments:** <https://github.com/willjobs/regulations-public-comments>
- **R regulationsgov:** <https://github.com/q-w-a/regulationsgov>
- **Regulations.gov comment scraper:**
<https://github.com/upenndigitalscholarship/regulations-gov-comment-scraper>
- **Sunlight regulations scraper:** <https://github.com/sunlightlabs/regulations-scraper>
- **eCFR analyzer:** <https://github.com/sam-berry/ecfr-analyzer>
- **eCFR analyzer fork:** <https://github.com/JohnDDuncanIII/ecfr-analyzer>
- **eCFR Markdown:** <https://github.com/AlextheYounga/ecfr>
- **CFPB regulations parser:** <https://github.com/cfpb/regulations-parser>

- **Regulatory data repository:** https://github.com/regulatorystudies/regulatory_data_repository
- **USAspending API repo:** <https://github.com/fedspendingtransparency/usaspending-api>
- **USA spending Python:** https://github.com/codeforamerica/usa_spending_python
- **awesome procurement data:** <https://github.com/makegov/awesome-procurement-data>
- **pysam:** <https://github.com/jpleger/pysam>
- **sam-search:** <https://github.com/MindPetal/sam-search>
- **govbizops:** <https://github.com/pretorin-ai/govbizops>
- **OpenSAM:** <https://github.com/akshayakula/OpenSAM>
- **capture MCP server:** <https://github.com/blencorp/capture-mcp-server>
- **990 XML reader / IRSx:** <https://github.com/jsfenfen/990-xml-reader>
- **Giving Tuesday 990 parser:** <https://github.com/Giving-Tuesday/form-990-xml-parser>
- **Giving Tuesday 990 mapper:** <https://github.com/Giving-Tuesday/form-990-xml-mapper>
- **IRS 990 efile R:** <https://github.com/Nonprofit-Open-Data-Collective/irs990efile>
- **Charity Navigator IRS 990:** <https://github.com/CharityNavigator/irs990>
- **Charity Navigator 990 long:** https://github.com/CharityNavigator/990_long
- **CourtListener repo:** <https://github.com/freelawproject/courtlistener>
- **CourtListener API client:** <https://github.com/freelawproject/courtlistener-api-client>
- **lawpy:** <https://github.com/paultopia/lawpy>
- **OpenSanctions:** <https://github.com/opensanctions/opensanctions>
- **OpenSanctions API examples:** <https://github.com/opensanctions/api-examples>
- **OFAC Go library:** <https://github.com/cardonator/ofac>
- **Sanction list search:** https://github.com/perchrh/sanction_list_search
- **Historical OFAC panel:** <https://github.com/jeffallen13/ofac>
- **opyncorporates:** <https://github.com/pjryan126/opyncorporates>

- **OpenCorporates Node:** <https://github.com/mikemaccana/opencorporates>
- **pygleif:** <https://github.com/ggravlingen/pygleif>
- **leipy:** <https://github.com/emredjan/leipy>
- **GLEIF MCP server:** <https://github.com/olgasafonova/gleif-mcp-server>
- **WeThePeople:** <https://github.com/Obelus-Labs-LLC/WeThePeople>
- **US gov open data MCP:** <https://github.com/lzinga/us-gov-open-data-mcp>
- **FedDocMCP:** <https://github.com/n8daniels/FedDocMCP>

21.3 Claude / Anthropic financial skills and agents

- **Anthropic financial services repo:** <https://github.com/anthropics/financial-services>
- **Anthropic finance agents announcement:**
<https://www.anthropic.com/news/finance-agents>
- **Anthropic financial services announcement:**
<https://www.anthropic.com/news/advancing-claude-for-financial-services>
- **Claude Cowork financial-services plugin install guide:**
<https://support.claude.com/en/articles/13851150-install-financial-services-plugins-for-cowork>
- **Anthropic skills repo:** <https://github.com/anthropics/skills>
- **Anthropic agent skills engineering post:**
<https://www.anthropic.com/engineering/equipping-agents-for-the-real-world-with-agent-skills>
- **Anthropic docs:** <https://docs.anthropic.com/>
- **Claude docs:** <https://docs.claude.com/>

21.4 OpenAI Codex development workflow

- **Codex web docs:** <https://developers.openai.com/codex/cloud>
- **Codex GitHub integration docs:**
<https://developers.openai.com/codex/integrations/github>
- **Codex GitHub Action docs:** <https://developers.openai.com/codex/github-action>
- **Codex app docs:** <https://developers.openai.com/codex/app>
- **OpenAI Codex CLI repo:** <https://github.com/openai/codex>

21.5 Data lake, warehouse, graph, search, and scaling

- **Apache Iceberg:** <https://iceberg.apache.org/>

- **Apache Iceberg docs:** <https://iceberg.apache.org/docs/latest/>
- **Delta Lake:** <https://delta.io/>
- **Apache Hudi:** <https://hudi.apache.org/>
- **Apache Parquet:** <https://parquet.apache.org/>
- **DuckDB:** <https://duckdb.org/>
- **DuckDB GitHub:** <https://github.com/duckdb/duckdb>
- **MotherDuck:** <https://motherduck.com/>
- **Trino:** <https://trino.io/>
- **PrestoDB:** <https://prestodb.io/>
- **AWS Athena:** <https://aws.amazon.com/athena/>
- **BigQuery:** <https://cloud.google.com/bigquery>
- **BigQuery pricing:** <https://cloud.google.com/bigquery/pricing>
- **Snowflake:** <https://www.snowflake.com/en/>
- **Snowflake pricing:** <https://www.snowflake.com/en/pricing-options/>
- **Databricks:** <https://www.databricks.com/>
- **ClickHouse:** <https://clickhouse.com/>
- **ClickHouse docs:** <https://clickhouse.com/docs/intro>
- **Postgres:** <https://www.postgresql.org/>
- **pgvector:** <https://github.com/pgvector/pgvector>
- **OpenSearch:** <https://opensearch.org/>
- **OpenSearch vector search docs:** <https://docs.opensearch.org/latest/vector-search/>
- **Elasticsearch:** <https://www.elastic.co/elasticsearch>
- **Qdrant:** <https://qdrant.tech/>
- **Weaviate:** <https://weaviate.io/>
- **Pinecone:** <https://www.pinecone.io/>
- **Neo4j:** <https://neo4j.com/>
- **ArangoDB:** <https://arangodb.com/>

- **Amazon Neptune:** <https://aws.amazon.com/neptune/>
- **JanusGraph:** <https://janusgraph.org/>
- **AWS S3:** <https://aws.amazon.com/s3/>
- **Google Cloud Storage:** <https://cloud.google.com/storage>
- **Azure Blob Storage:** <https://azure.microsoft.com/en-us/products/storage/blobs>
- **Cloudflare R2:** <https://www.cloudflare.com/developer-platform/products/r2/>
- **MinIO:** <https://min.io/>
- **Dagster:** <https://dagster.io/>
- **Prefect:** <https://www.prefect.io/>
- **Apache Airflow:** <https://airflow.apache.org/>
- **dbt:** <https://www.getdbt.com/>
- **Great Expectations:** <https://greatexpectations.io/>
- **Splink:** <https://github.com/moj-analytical-services/splink>
- **Dedupe:** <https://github.com/dedupeio/dedupe>
- **OpenRefine:** <https://www.openrefine.org/>

21.6 Market, financial, company, and research data providers

- **Massive / Polygon docs:** <https://massive.com/docs>
- **Polygon docs:** <https://polygon.io/docs>
- **Finnhub API:** <https://finnhub.io/docs/api>
- **Financial Modeling Prep docs:**
<https://site.financialmodelingprep.com/developer/docs>
- **Nasdaq Data Link docs:** <https://docs.data.nasdaq.com/>
- **Alpha Vantage docs:** <https://www.alphavantage.co/documentation/>
- **Crunchbase data docs:** <https://data.crunchbase.com/docs/welcome-to-crunchbase-data>
- **Dealroom API:** <https://dealroom.co/api>
- **PitchBook:** <https://pitchbook.com/>
- **CB Insights:** <https://www.cbinsights.com/>

- **Tracxn:** <https://tracxn.com/>
- **Preqin:** <https://www.preqin.com/>
- **S&P Capital IQ / Market Intelligence:**
<https://www.spglobal.com/marketintelligence/en/>
- **FactSet:** <https://www.factset.com/>
- **AlphaSense:** <https://www.alpha-sense.com/>
- **OpenAlex API:** <https://developers.openalex.org/api-reference/introduction>
- **NIH RePORTER API:** <https://api.reporter.nih.gov/>
- **USPTO data APIs:** <https://data.uspto.gov/>
- **PatentsView APIs:** <https://patentsview.org/apis/purpose>
- **Crossref REST API:** <https://www.crossref.org/documentation/retrieve-metadata/rest-api/>
- **Semantic Scholar API:** <https://www.semanticscholar.org/product/api>

21.7 OSINT, investigative, procurement, and red-flag methodology

- **FollowTheMoney schema:** <https://github.com/opensanctions/followthemoney>
- **FollowTheMoney docs:** <https://followthemoney.tech/>
- **OCCRP Aleph:** <https://aleph.occrp.org/>
- **Aleph GitHub:** <https://github.com/alephdata/aleph>
- **OpenAleph:** <https://openaleph.org/>
- **OpenAleph GitHub:** <https://github.com/openaleph/openaleph>
- **OCCRP:** <https://www.occrp.org/>
- **Follow the Money handbook:** <https://exposingtheinvisible.org/en/guides/follow-the-money-handbook/>
- **ICIJ Offshore Leaks:** <https://offshoreleaks.icij.org/>
- **ICIJ Offshore Leaks reconciliation:** <https://offshoreleaks.icij.org/docs/reconciliation>
- **Bellingcat toolkit:** <https://bellingcat.gitbook.io/toolkit>
- **Bellingcat online investigations toolkit article:**
<https://www.bellingcat.com/resources/2024/09/24/bellingcat-online-investigations-toolkit/>
- **SpiderFoot:** <https://github.com/smicallef/spiderfoot>

- **SpiderFoot website:** <https://www.spiderfoot.net/>
- **OpenCTI:** <https://github.com/opencti-platform/opencti>
- **OpenCTI website:** <https://www.opencti.io/>
- **Open Contracting Data Standard:** <https://www.open-contracting.org/data-standard/>
- **Open Contracting red flags guide:**
<https://www.open-contracting.org/resources/red-flags-in-public-procurement-a-guide-to-using-data-to-detect-and-mitigate-risks/>
- **Cardinal red flags announcement:**
<https://www.open-contracting.org/2024/06/12/cardinal-an-open-source-library-to-calculate-public-procurement-red-flags/>
- **Cardinal repository:** <https://github.com/open-contracting/cardinal-rs>
- **World Bank procurement awards:**
<https://financesone.worldbank.org/procurement-contract-awards-summary-by-fiscal-year-and-supplier-country/DS01529>
- **UNGM:** <https://www.ungm.org/>
- **UNGM developer center:** <https://developer.ungm.org/>

21.8 Phase 2 global sources

- **UK Companies House developer portal:** <https://developer.company-information.service.gov.uk/>
- **Companies House public data API reference:** <https://developer-specs.company-information.service.gov.uk/companies-house-public-data-api/reference>
- **Companies House PSC API reference:** <https://developer-specs.company-information.service.gov.uk/companies-house-public-data-api/reference/persons-with-significant-control>
- **UK Contracts Finder:** <https://www.gov.uk/contracts-finder>
- **UK Find a Tender:** <https://www.find-tender.service.gov.uk/>
- **UK Electoral Commission:** <https://www.electoralcommission.org.uk/>
- **UK sanctions consolidated lists:**
<https://www.gov.uk/government/collections/financial-sanctions-regime-specific-consolidated-lists-and-releases>
- **EU Transparency Register:** https://commission.europa.eu/about/service-standards-and-principles/transparency/transparency-register_en
- **EU TED:** <https://ted.europa.eu/en/>

- **TED developer corner:** <https://ted.europa.eu/en/simap/developers-corner-for-reusers>
- **TED docs:** <https://docs.ted.europa.eu/home/index.html>
- **EU Financial Transparency System dataset:** <https://data.europa.eu/data/datasets/fts>
- **EU Sanctions Map:** <https://www.sanctionsmap.eu/>
- **CORDIS:** <https://cordis.europa.eu/>
- **Elections Canada political finance:** <https://www.elections.ca/content.aspx?section=fin&document=index&lang=e>
- **Canada lobbying open data:** <https://lobbycanada.gc.ca/en/open-data/>
- **CanadaBuys:** <https://canadabuys.canada.ca/>
- **Australian Electoral Commission Transparency Register:** https://www.aec.gov.au/parties_and_representatives/financial_disclosure/transparency-register/
- **Australian Government Register of Lobbyists:** <https://www.ag.gov.au/integrity/australian-government-register-lobbyists/information-lobbyists>
- **AusTender:** <https://www.tenders.gov.au/>
- **ASIC:** <https://asic.gov.au/>
- **Brazil Portal da Transparencia API:** <https://portaldatransparencia.gov.br/api-dados>
- **Brazil Compras.gov.br open data:** <https://www.gov.br/compras/pt-br/cidadao/compras-publicas-dados-abertos>
- **Brazil Camara Dados Abertos API:** <https://dadosabertos.camara.leg.br/swagger/api.html>
- **Brazil Senado Dados Abertos API:** <https://legis.senado.leg.br/dadosabertos/api-docs/swagger-ui/index.html>
- **Brazil CVM Dados Abertos:** <https://dados.cvm.gov.br/>
- **Brazil TSE Dados Abertos:** <https://dadosabertos.tse.jus.br/>
- **Brazil e-Agendas:** <https://eagendas.cgu.gov.br/>
- **Brazil API catalog:** <https://www.gov.br/conecta/catalogo/apis/api-portal-de-dados-abertos>

21.9 News, events, assets, report tooling, and UI libraries

- **GDELT:** <https://www.gdeltproject.org/>
- **GDELT data:** <https://www.gdeltproject.org/data.html>
- **Media Cloud:** <https://www.mediacloud.org/>
- **Media Cloud API support:** <https://mediacloud.org/support/api>
- **FAA aircraft registry:**
https://www.faa.gov/licenses_certificates/aircraft_certification/aircraft_registry
- **FAA releasable aircraft download:**
https://www.faa.gov/licenses_certificates/aircraft_certification/aircraft_registry/releasable_aircraft_download
- **FCC public access database downloads:** <https://www.fcc.gov/wireless/data/public-access-files-database-downloads>
- **FCC database search:** <https://www.fcc.gov/licensing-databases/search-fcc-databases>
- **Regrid parcel API:** <https://regrid.com/api>
- **USGS parcel viewer:**
<https://www.usgs.gov/core-science-systems/ngp/tnm-corps/authoritative-source-parcel-viewer>
- **NOAA vessel/related source:** <https://www.fisheries.noaa.gov/inport/item/20701>
- **USCG PSIX:** <https://cgmix.uscg.mil/psix/>
- **ReportLab docs:** <https://www.reportlab.com/dev/docs/>
- **python-docx docs:** <https://python-docx.readthedocs.io/>
- **python-pptx docs:** <https://python-pptx.readthedocs.io/>
- **openpyxl docs:** <https://openpyxl.readthedocs.io/>
- **pandas docs:** <https://pandas.pydata.org/docs/>
- **React Flow:** <https://reactflow.dev/>
- **TanStack Query:** <https://tanstack.com/query/latest>
- **TanStack Table:** <https://tanstack.com/table/latest>
- **AG Grid:** <https://www.ag-grid.com/>
- **D3.js:** <https://d3js.org/>

22 Appendix B - Source Contract Template

```
source: sec_edgar
display_name: SEC EDGAR
phase: phase_1_us_mvp
official: true
base_url: https://data.sec.gov
auth:
  type: none_or_api_key
rate_limit:
  requests_per_second: 10
raw_storage: required
normalization: required
evidence_required: true
refresh:
  discovery: hourly
  historical_backfill: batch
  full_refresh: manual_only
outputs:
  entities:
    - Company
    - Filing
    - Person
    - Security
  relationships:
    - FILED
    - OFFICER_OF
    - HOLDS
    - INSIDER_TRANSACTION
compliance_notes:
  - Follow source terms and fair access policies.
  - Store raw documents with hashes.
  - Do not overload source API.
quality_checks:
  - missing_ids
  - duplicate_source_records
  - parser_errors
  - stale_data
  - broken_document_urls
```

23 Appendix C - Definition of Done

A module is complete only when:

- Feature is implemented behind permissions.
- API contract is documented.
- UI is usable by an analyst.
- Unit tests pass.
- Integration tests exist or are explicitly deferred.
- Audit events are emitted.
- Source/evidence references are attached where relevant.
- Error states are handled.
- Admin visibility exists for operational functions.
- Security review is completed for sensitive features.
- Documentation is written.
- Acceptance criteria are satisfied.